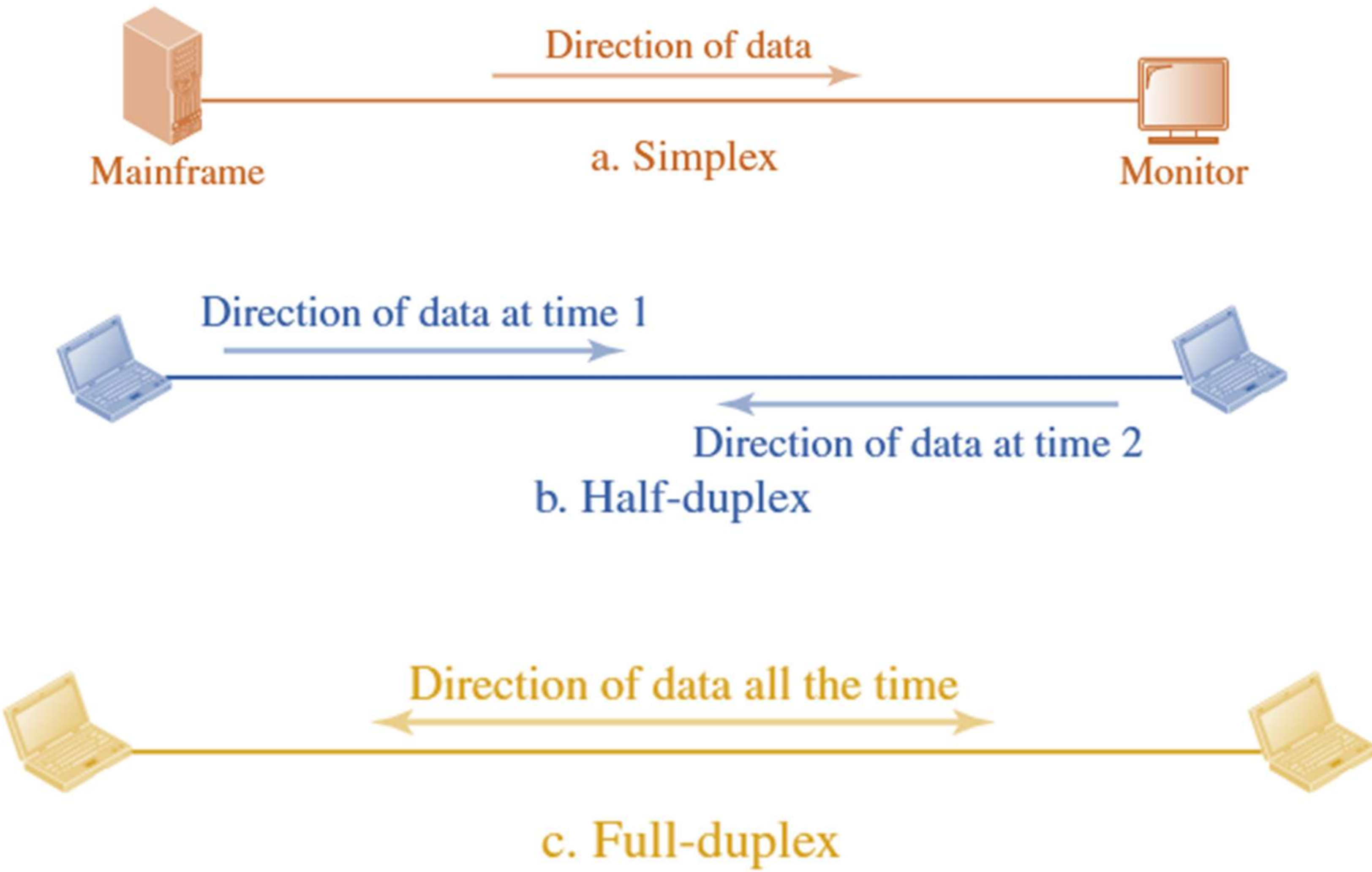
Module No. 2	Physical Layer
Guided media, Unguided media, Network Topologies, Analog and digital communication. Encoding mechanisms, Packet Switching, Circuit Switching.	

Network Topologies

Data Flow

Communication between two devices can be

- Simplex
- Half-duplex
- Full-duplex


Direction of data
Mainframe
a. Simplex
Monitor
Direction of data at time 1
Direction of data at time 2
b. Half-duplex
Direction of data all the time
c. Full-duplex

Types of Connection

Two types of connections:

- Point-to-Point

a. Point-to-point

- Multipoint or Multidrop

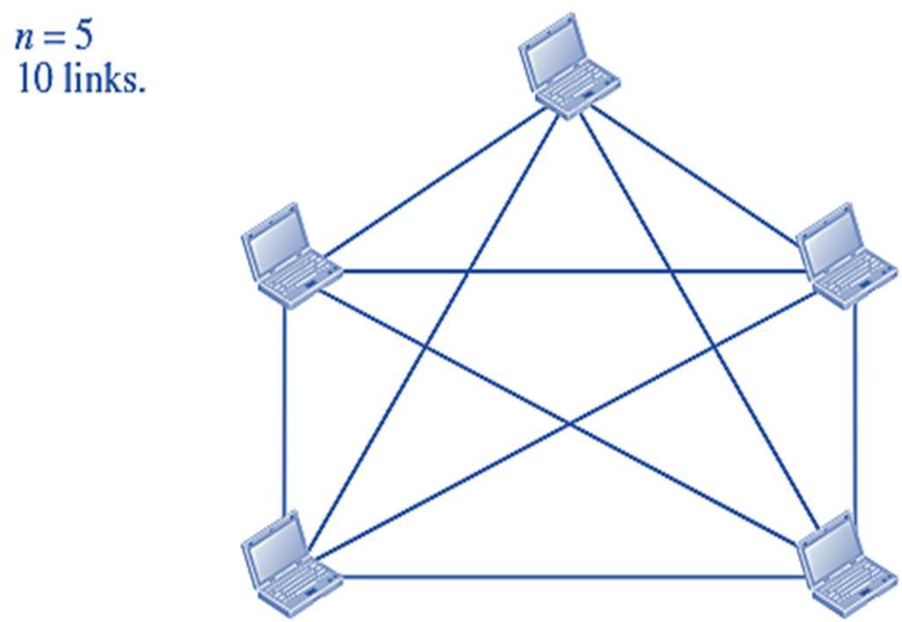
b. Multipoint

In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.

- If several devices can use the link simultaneously, it is a spatially shared connection.
- If users must take turns, it is a timeshared connection.

Mesh Topology

- Has a dedicated point-to-point link to every other device.
- If the number of physical links in a fully connected mesh network with n nodes, need **n (n – 1) / 2 duplex-mode link**.
- To accommodate that many links, every device on the network must have **n – 1 input/output (I/O) ports** to be connected to the other n – 1 stations.



Several advantages over other network topologies.

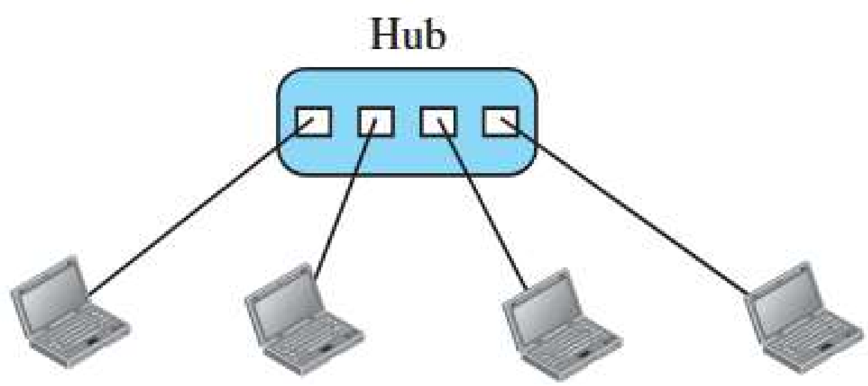
- First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
- Second, a mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.
- Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
- Finally, point-to-point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution

Disadvantages of a mesh:

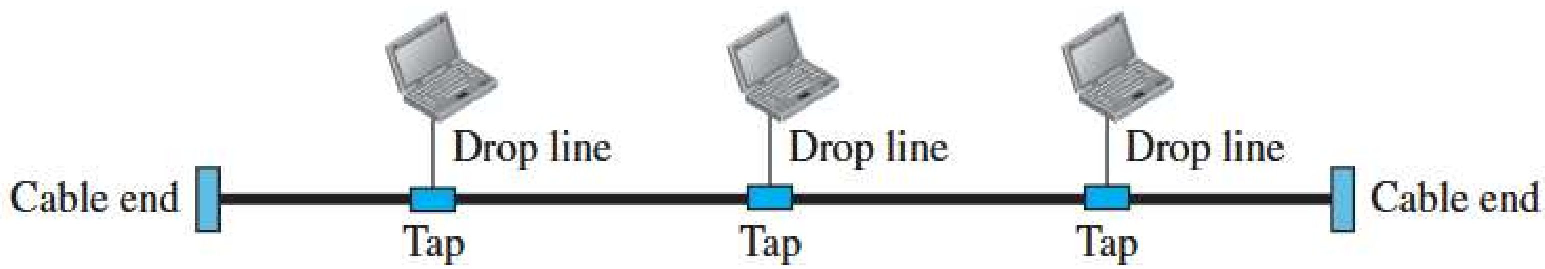
- High cost of amount of cabling and the number of I/O ports required.
- Because every device must be connected to every other device, installation and reconnection are difficult. The sheer bulk of the wiring can be greater than the available space can accommodate.
- The hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

Star Topology

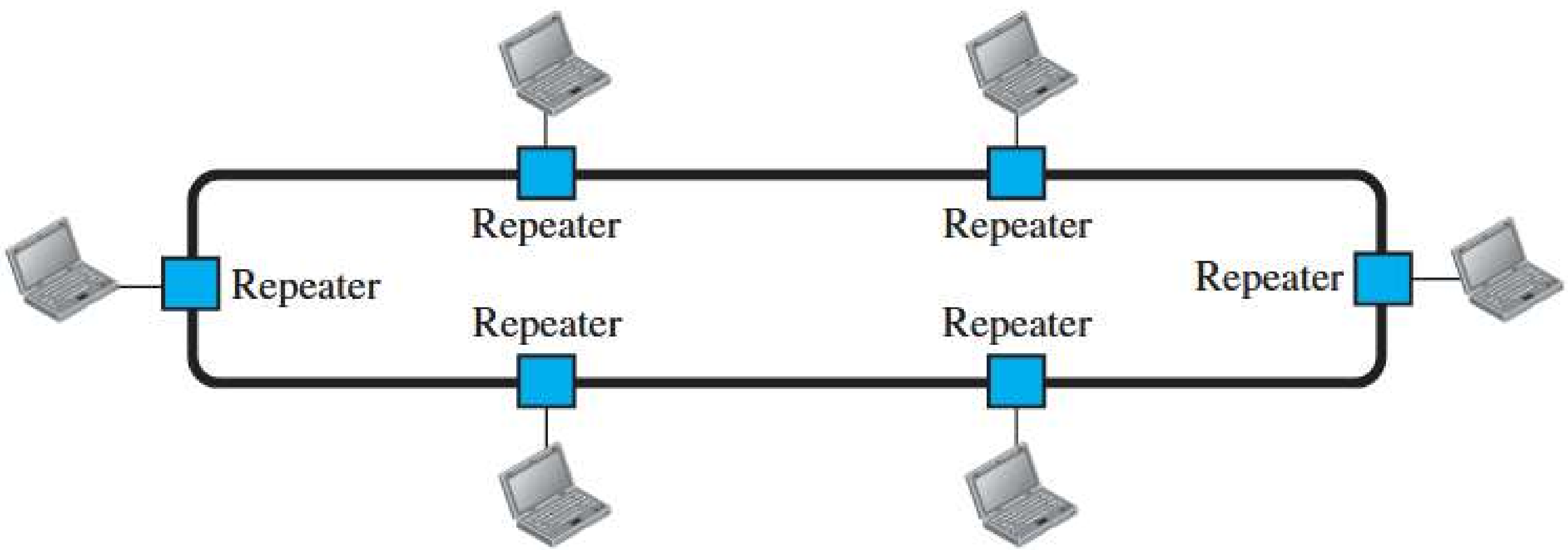
In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.



Bus Topology



Ring Topology



Analog and Digital Data

Analog data refers to information that is continuous

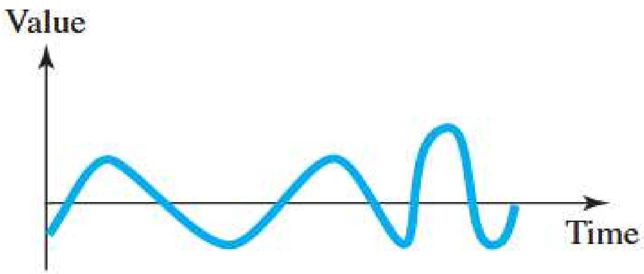
- Sounds made by a human voice, take on continuous values. When someone speaks, an analog wave is created in the air.

Digital data refers to information that has discrete states

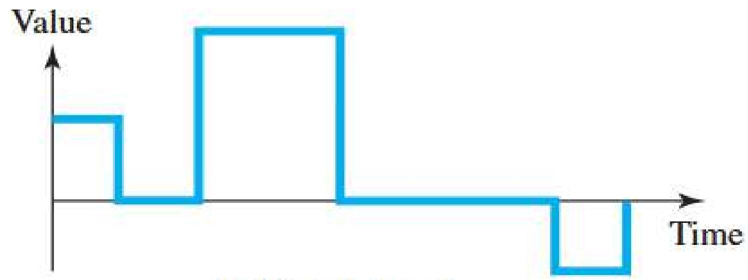
- Data are stored in computer memory in the form of 0s and 1s.

An Analog signal has infinitely many levels of intensity over a period of time.

A Digital signal, on the other hand, can have only a limited number of defined values.



a. Analog signal



b. Digital signal

Analog Signals

- periodic
- Generally used in Data communications
- Classified as simple or composite.
- Simple periodic signal is called Sine Wave; Cannot be decomposed further
- Aperiodic

A sine wave can be represented by three parameters:

- period
- peak amplitude
- phase

Peak Amplitude

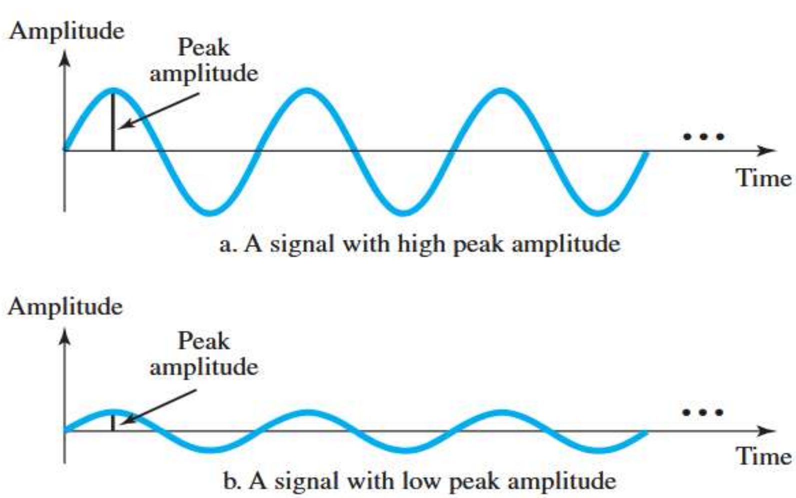
The peak amplitude of a signal is the absolute value of its highest intensity.
For electrical signals, peak amplitude is normally measured in volts.

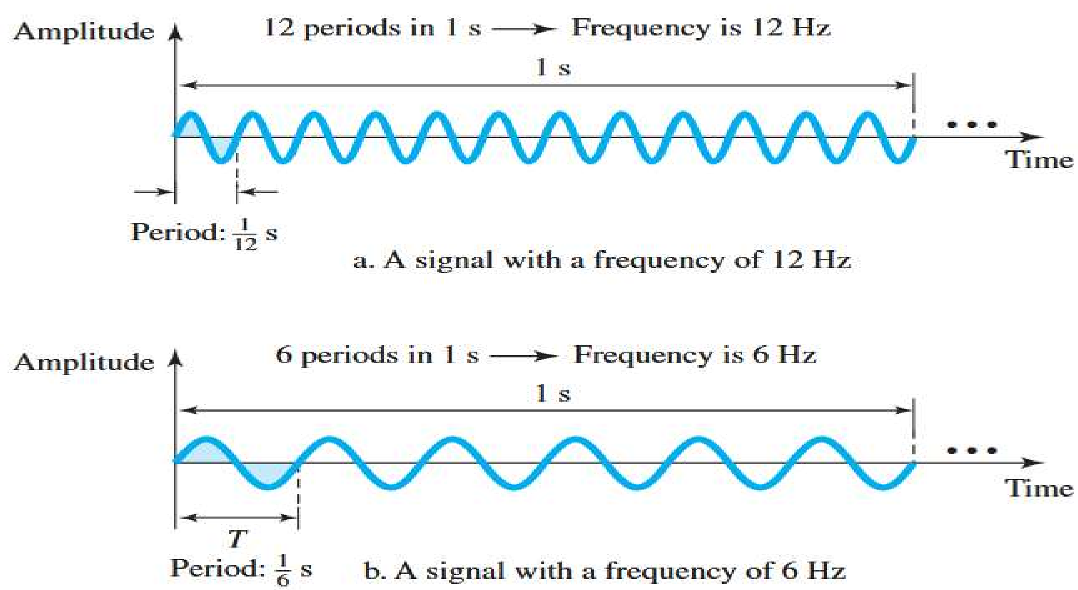
Period and Frequency

Period refers to the amount of time, in seconds, a signal needs to complete 1 cycle.
Frequency refers to the number of periods in 1 s.
Period is the inverse of frequency, and frequency is the inverse of period.
Period is formally expressed in seconds. Frequency is formally expressed in Hertz (Hz), which is cycle per second.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Two signals with the same phase and frequency, but different amplitudes



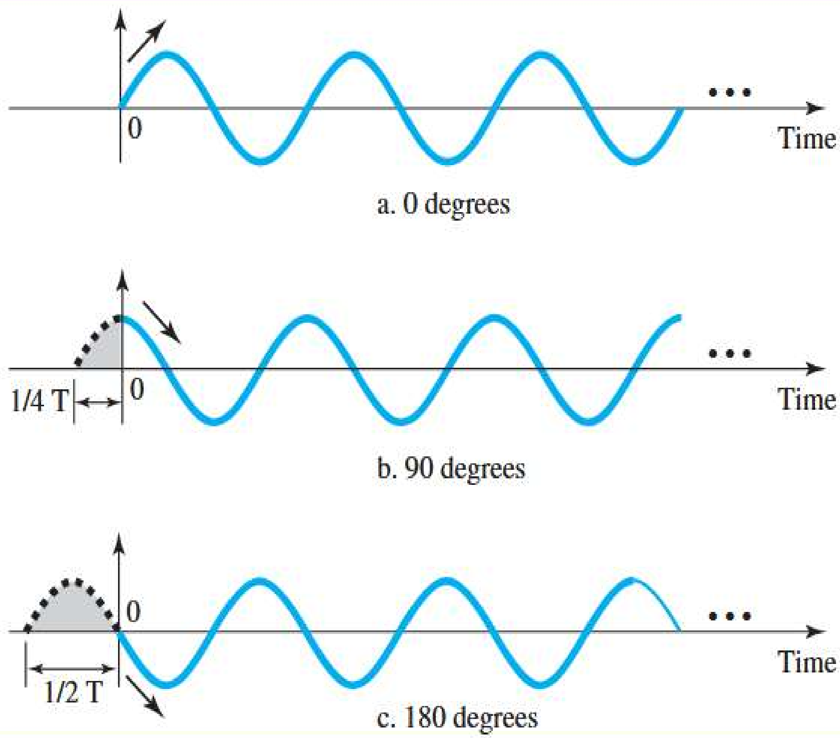


Phase

The term phase describes the position / the amount of the shift of the waveform relative to time 0.

Phase is measured in degrees or radians (360° is 2π rad).

Three sine waves with the same amplitude and frequency, but different phases



A sine wave is offset 1/6 cycle with respect to time 0. What is its phase in degrees and radians?

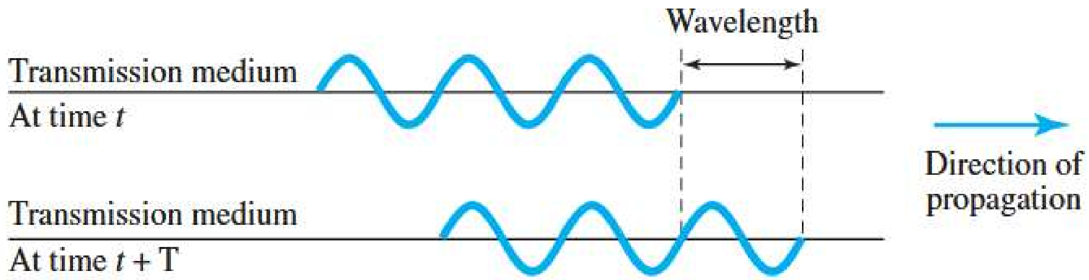
It is known that 1 complete cycle is 360°.
Therefore, 1/6 cycle is

$$\frac{1}{6} \times 360 = 60^\circ$$

$$= 60 \times \frac{2\pi}{360} \text{ rad} = \frac{\pi}{3} \text{ rad} = 1.046 \text{ rad}$$

Wavelength

- The wavelength is the distance a simple signal can travel in one period.
- Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium



Since period and frequency are related to each other, wavelength is represented by **λ**, propagation speed by **c** (speed of light), and frequency by **f**, we get

$$\textbf{Wavelength} = (\textbf{propagation speed}) \times \textbf{period} = \frac{\textbf{propagation speed}}{\textbf{frequency}}$$

$$\lambda = \frac{c}{f}$$

The wavelength is normally measured in micrometers (microns) instead of meters.

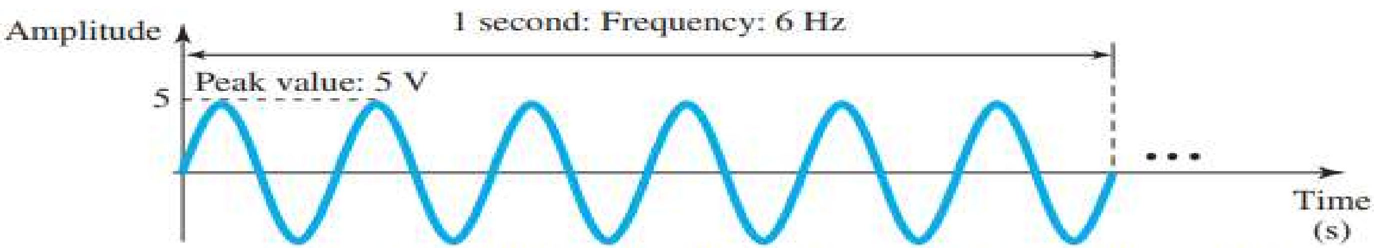
The light is propagated with a speed of 3×10^8 m/s. The wavelength of red light (frequency = 4×10^{14}) in air is

$$\lambda = \frac{c}{f} = \frac{3\times 10^8}{4\times 10^{14}} = 0.75 \times 10^{-6} \text{ m} = 0.75 \ \mu\text{m}$$

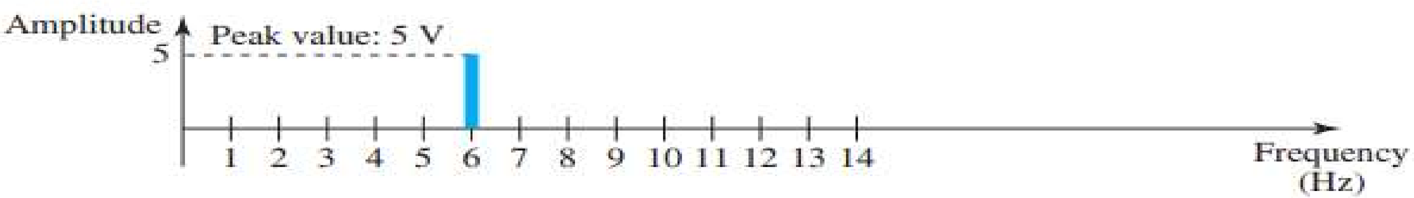
In a coaxial or fiber-optic cable, however, the wavelength is shorter (0.5 μm) because the propagation speed in the cable is decreased.

Time and Frequency Domains

- A sine wave is comprehensively defined by its amplitude, frequency, and phase.
- The time-domain plot shows changes in signal amplitude with respect to time. Phase is not explicitly shown on a time-domain plot.
- A frequency-domain plot is concerned with only the peak value and the frequency. Changes of amplitude during one period are not shown.



a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



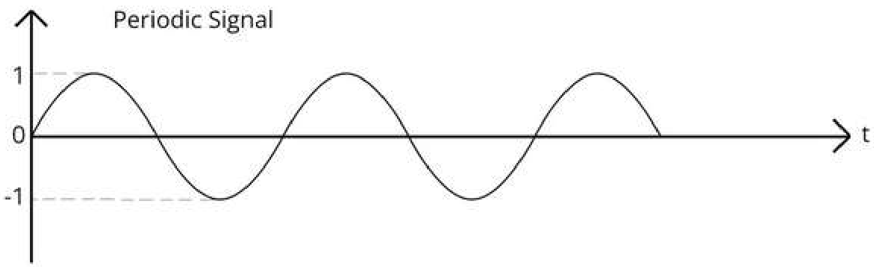
b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

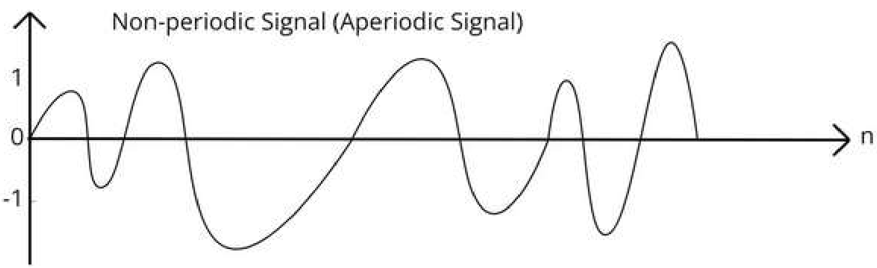
- The advantage of the frequency domain is that the values of the frequency and peak amplitude are clearly seen.
- A complete sine wave is represented by one spike.
- The position of the spike shows the frequency; its height shows the peak amplitude.

Composite Signals

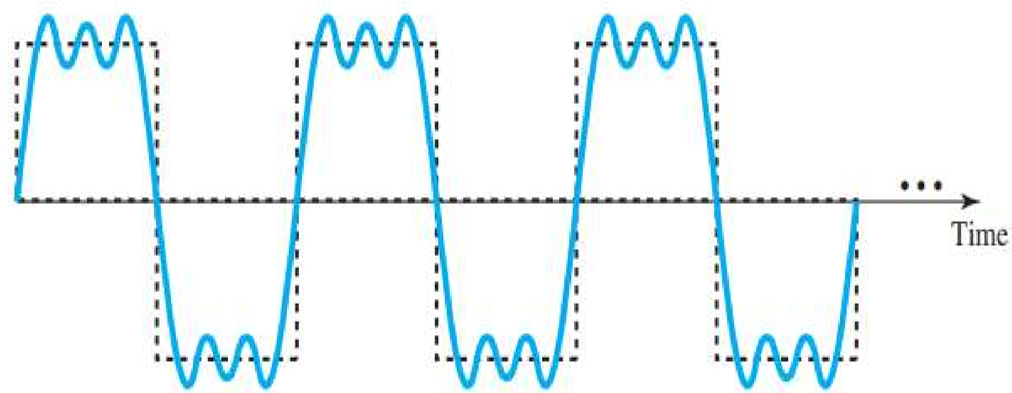
- A single-frequency sine wave is not useful in data communications; but a composite signal, a signal made of many simple sine waves, is needed.
- According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases. ***French mathematician Jean-Baptiste Fourier*

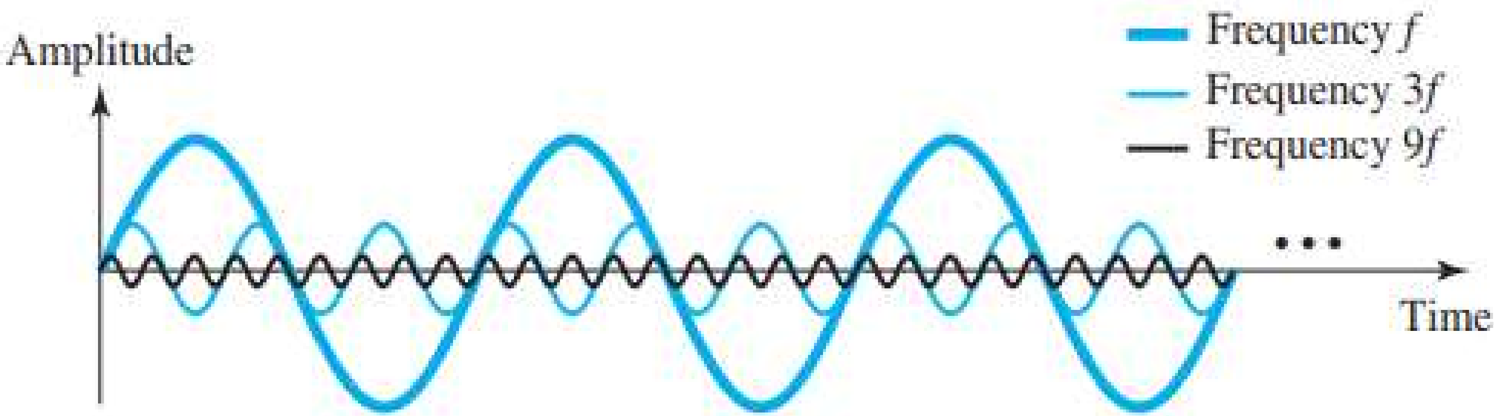
  - If the composite signal is periodic, the decomposition gives a series of signals with discrete frequencies;
  - If the composite signal is non-periodic, the decomposition gives a combination of sine waves with continuous frequencies.



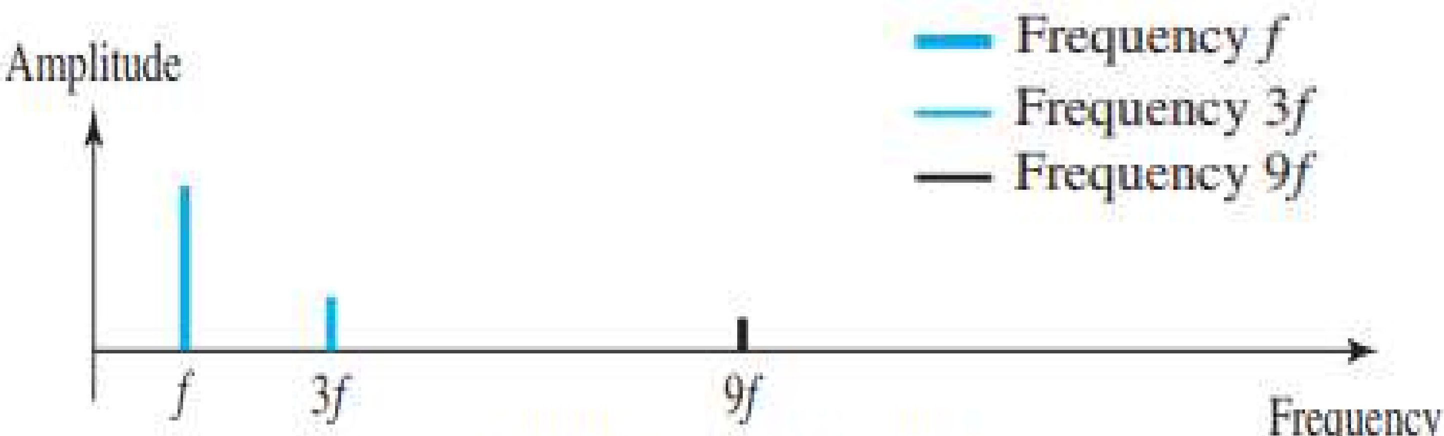


A composite periodic signal





a. Time-domain decomposition of a composite signal



b. Frequency-domain decomposition of the composite signal

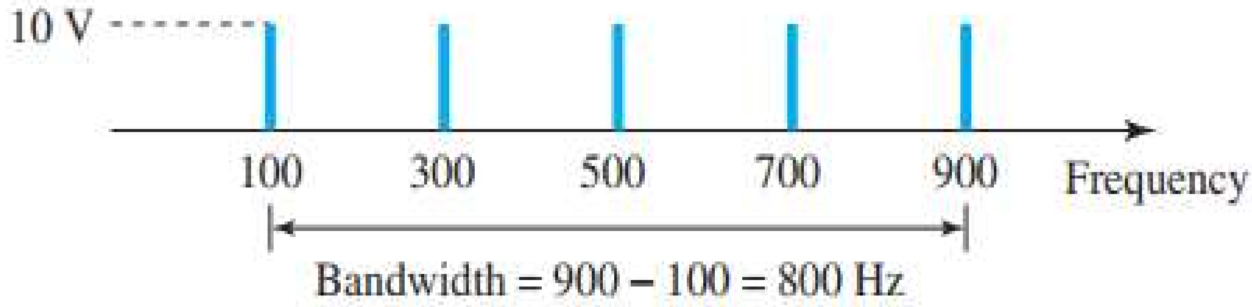
Bandwidth

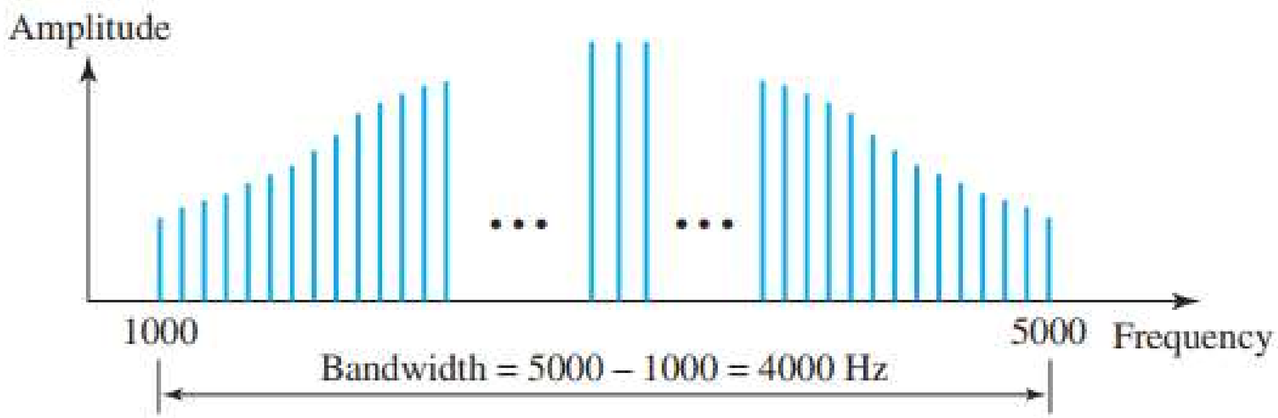
The range of frequencies contained in a composite signal is its bandwidth. The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

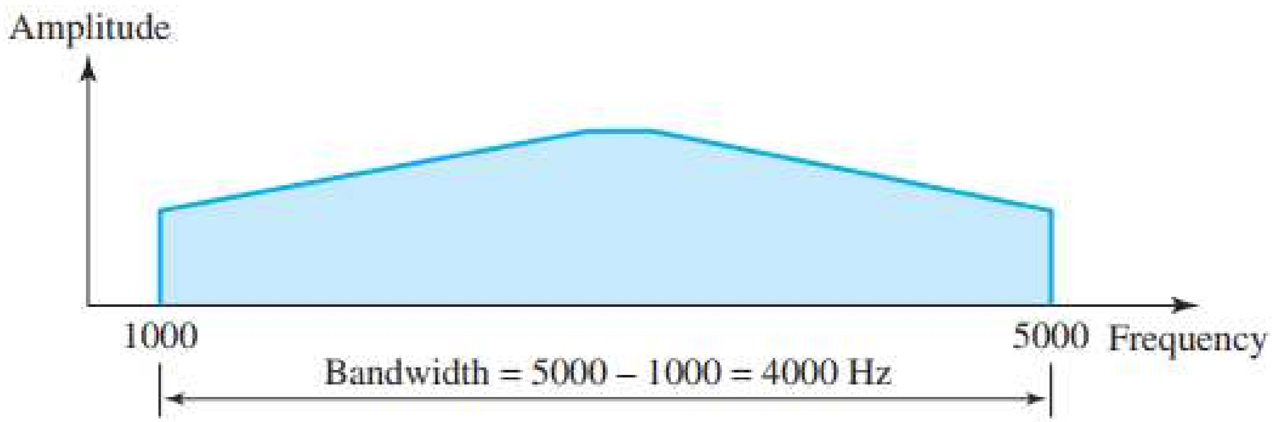
Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$





a. Bandwidth of a periodic signal



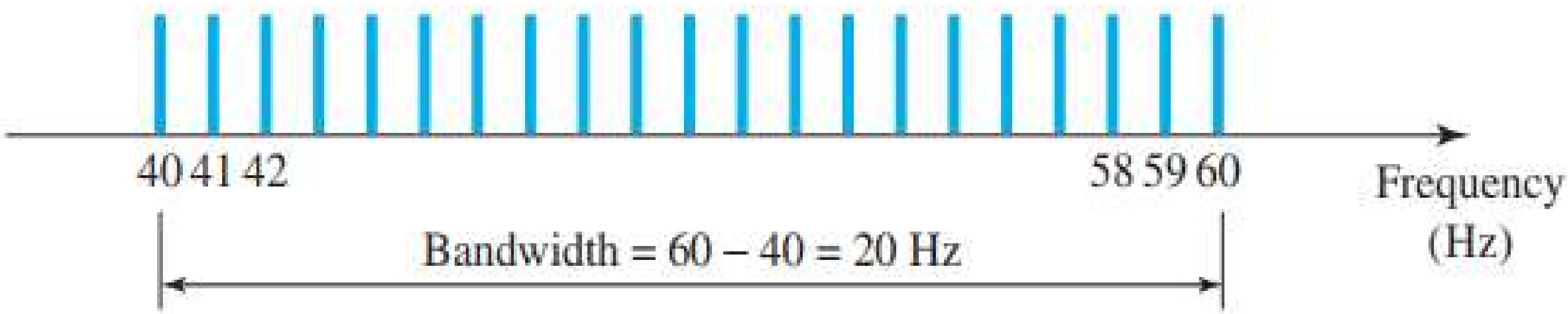
b. Bandwidth of a nonperiodic signal

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

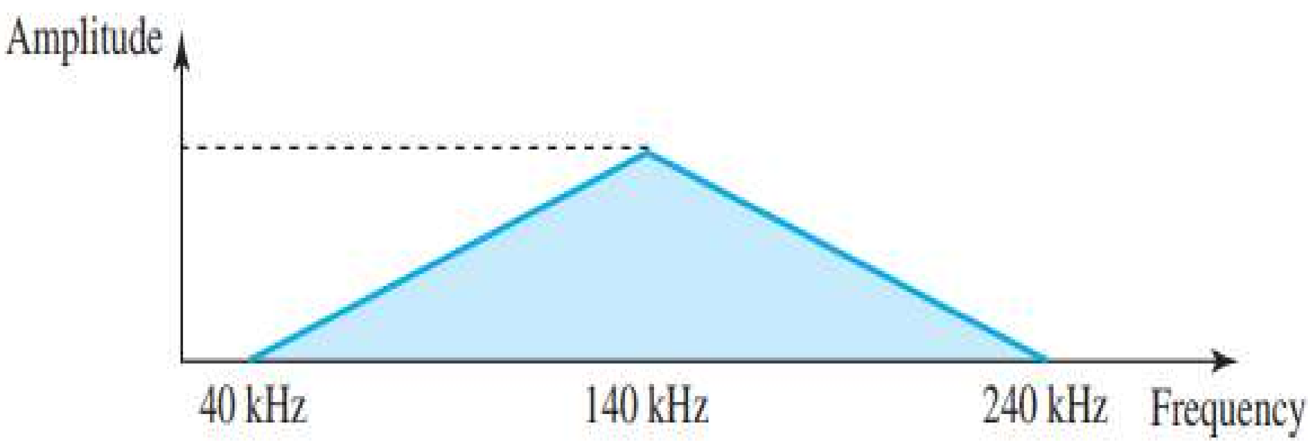
$$B = f_h - f_l \longrightarrow 20 = 60 - f_l \longrightarrow f_l = 60 - 20 = 40 \text{ Hz}$$



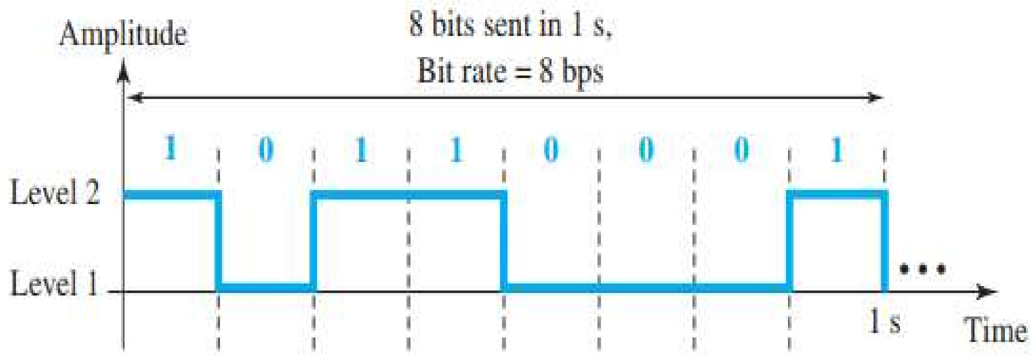
A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

$$\text{Lower Frequency} = f_c - \frac{BW}{2} = 140\,\text{kHz} - 100\,\text{kHz} = 40\,\text{kHz}$$

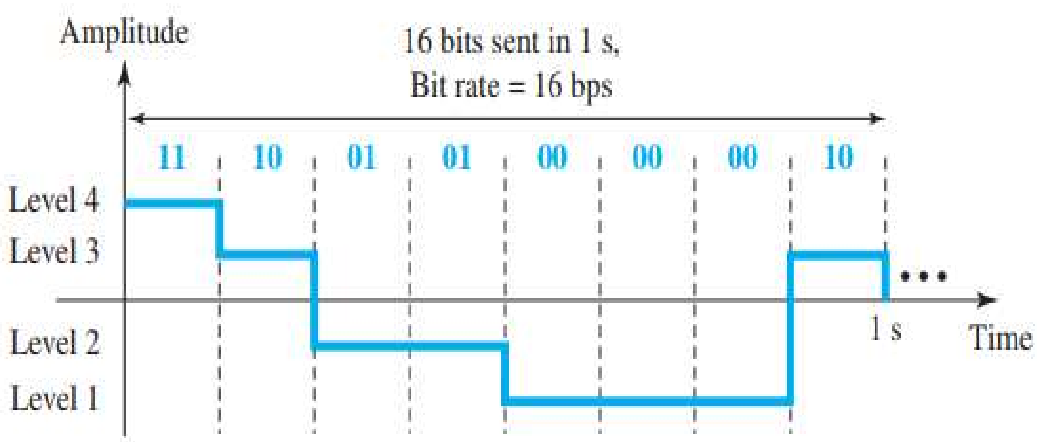
$$\text{Upper Frequency} = f_c + \frac{BW}{2} = 140\,\text{kHz} + 100\,\text{kHz} = 240\,\text{kHz}$$



DIGITAL SIGNALS



a. A digital signal with two levels



b. A digital signal with four levels

- In addition to being represented by an analog signal, information can also be represented by a digital signal.
- A 1 can be encoded as a positive voltage and a 0 as zero voltage.
- A digital signal can have more than two levels.
- In general, if a signal has **L levels**, each level needs **$\log_2 L$ bits**.

Bit Rate

- Most digital signals are non-periodic, and thus period and frequency are not appropriate characteristics.
- The term, bit rate, is used to describe digital signals. The bit rate is the number of bits sent in 1s, expressed in bits per second (bps).

You need to download **100 text pages per second** over a communication channel. Each page contains:

- **25 lines per page**
- **80 characters per line**
- Each character is encoded using **ASCII (8 bits per character)**

Question:

What is the required bit rate of the channel to support this download speed?

Each page consists of 25 lines, and each line contains 80 characters. So, the total number of characters on one page is:

$$25 \times 80 = 2,000 \text{ characters}$$

Since each character is encoded using ASCII, which uses 8 bits per character, the number of bits per page is:

$$2,000 \times 8 = 16,000 \text{ bits}$$

At a rate of 100 pages per second, the total number of bits transmitted per second is:

$$16,000 \times 100 = \boxed{1,600,000 \text{ bps} = 1.6 \text{ Mbps}}$$

Bit Length The bit length is the distance **one bit** occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

or

$$\text{Bit length} = 1 / (\text{bit rate})$$

Transmission of Digital Signals

- A digital signal can be transmitted by one of two different approaches:
  - Baseband transmission or
  - Broadband transmission

- **Baseband transmission** - Sending a digital signal over a channel without changing it to an analog signal.
  - ✓ Baseband transmission requires that we have a **low-pass channel**, a channel with a bandwidth that starts from zero.
  - ✓ This is the case if we have a dedicated medium with a bandwidth constituting only one channel.
  - ✓ In baseband transmission, the required bandwidth is proportional to the bit rate;
  - ✓ To send bits faster, more bandwidth is needed.

Bit Rate	*Harmonic 1*	*Harmonics 1, 3*	*Harmonics 1, 3, 5*
$n = 1$ kbps	$B = 500$ Hz	$B = 1.5$ kHz	$B = 2.5$ kHz
$n = 10$ kbps	$B = 5$ kHz	$B = 15$ kHz	$B = 25$ kHz
$n = 100$ kbps	$B = 50$ kHz	$B = 150$ kHz	$B = 250$ kHz

What is the required bandwidth of a low-pass channel if we need to send 1 Mbps by using baseband transmission?

Solution

The answer depends on the accuracy desired.

a. The minimum bandwidth, a rough approximation, is $B =$ bit rate /2, or 500 kHz. We need a low-pass channel with frequencies between 0 and 500 kHz.

b. A better result can be achieved by using the first and the third harmonics with the required bandwidth $B = 3 \times 500$ kHz $= 1.5$ MHz.

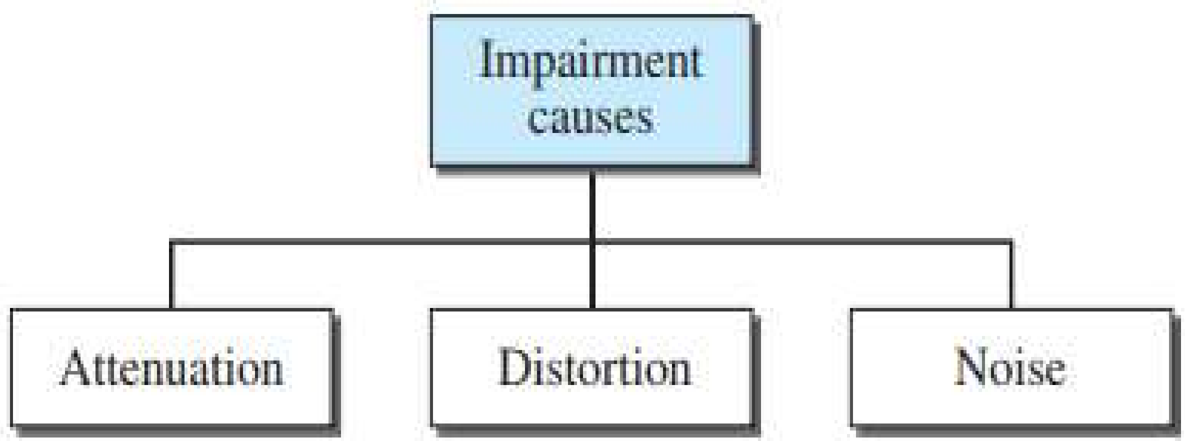
c. A still better result can be achieved by using the first, third, and fifth harmonics with $B = 5 \times 500$ kHz $= 2.5$ MHz.

- **Broadband transmission** or **modulation** - Changing the digital signal to an analog signal for transmission.
    - ✓ Modulation allows us to use a **bandpass channel**—a channel with a bandwidth that does not start from zero.
    - ✓ This type of channel is more available than a low-pass channel.

Bandpass channel

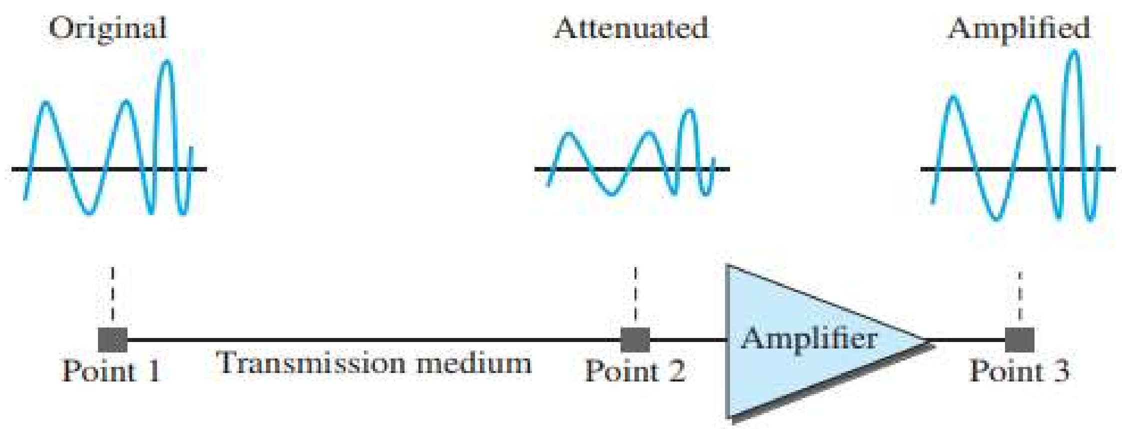
If the available channel is a bandpass channel, we cannot send the digital signal directly to the channel; we need to convert the digital signal to an analog signal before transmission.

TRANSMISSION IMPAIRMENT



Attenuation

- Attenuation means a loss of energy.
- When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- That is why a wire carrying electric signals gets warm, if not hot, after a while. Some of the electrical energy in the signal is converted to heat.
- To compensate for this loss, amplifiers are used to amplify the signal.



Decibel

- To show that a signal has lost or gained strength, engineers use the unit of the decibel.
- The decibel (dB) measures the relative strengths of two signals or one signal at two different points.
- The decibel is negative if a signal is attenuated and positive if a signal is amplified.

$$\mathrm{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

In this case, because power is proportional to the square of the voltage, the formula is

$$\mathbf{dB = 20 \log_{10} (V2/V1)}$$

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that $P_2 = \frac{1}{2} P_1$. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10 P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with −0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution

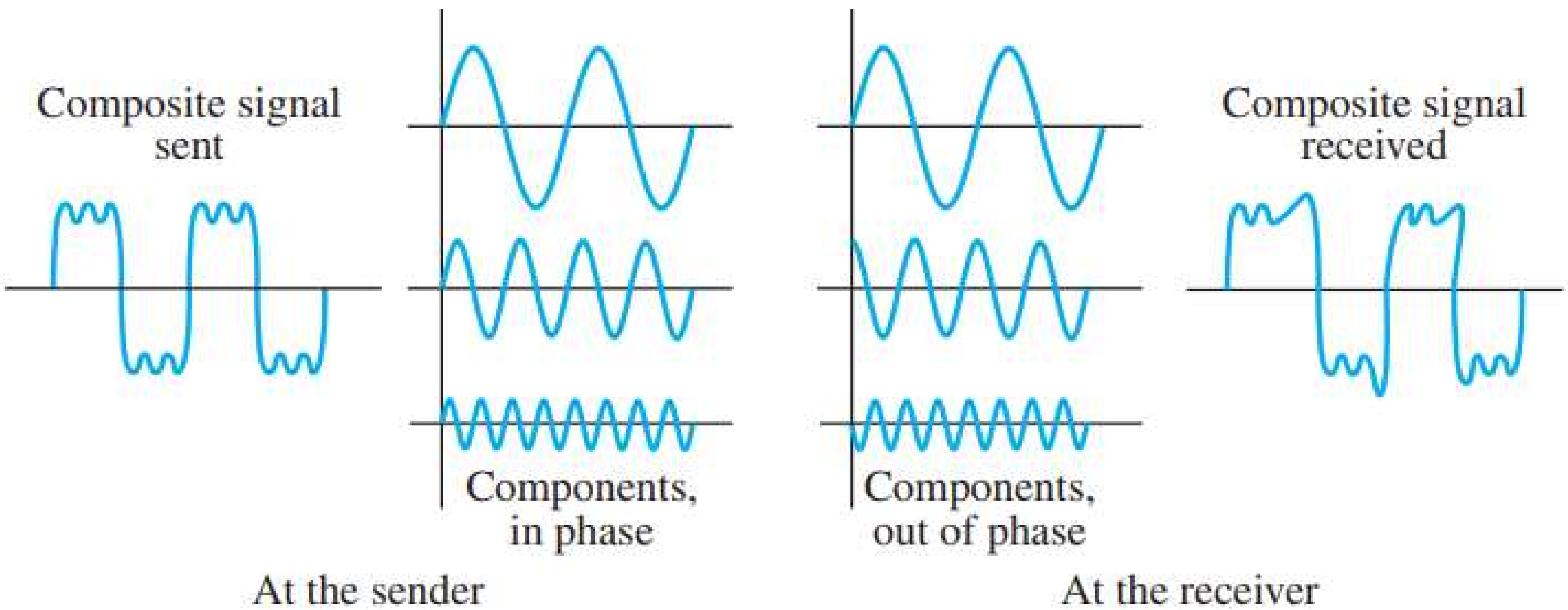
The loss in the cable in decibels is $5 \times (-0.3) = -1.5$ dB. We can calculate the power as

$$\mathbf{dB = 10 \log_{10} (P_2 / P_1) = -1.5 \quad \longrightarrow \quad (P_2 / P_1) = 10^{-0.15} = 0.71}$$

$$\mathbf{P_2 = 0.71 P_1 = 0.7 \times 2\ mW = 1.4\ mW}$$

Distortion

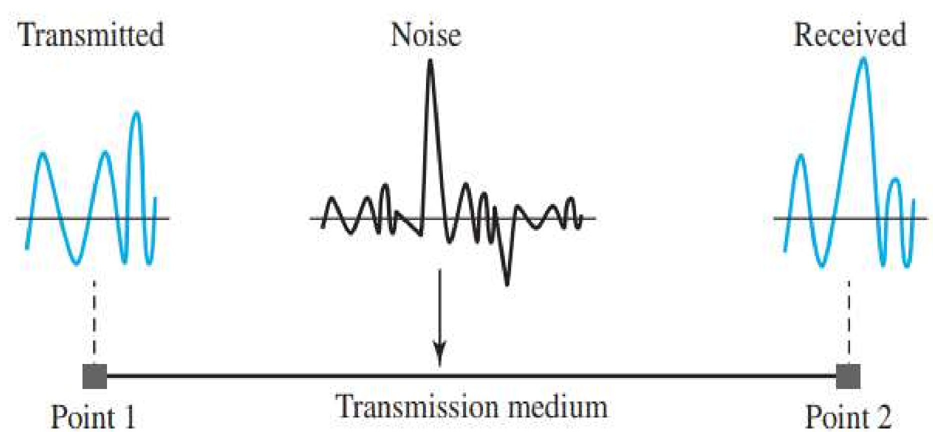
- Distortion means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies.
- HOW?
  - ✓ Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination.
  - ✓ Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.
  - ✓ In other words, signal components at the receiver have phases different from what they had at the sender.
  - ✓ The shape of the composite signal is therefore not the same.



Composite signal sent
Components, in phase
Components, out of phase
Composite signal received
At the sender
At the receiver

Noise

- Noise is another cause of impairment that may corrupt the signal.
  - Thermal noise is the random motion of electrons in a wire, which creates an extra signal not originally sent by the transmitter.
  - Induced noise comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna.
  - Crosstalk is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna.
  - Impulse noise is a spike (a signal with high energy in a very short time) that comes from power lines, lightning.



Signal-to-Noise Ratio (SNR)

The signal-to-noise ratio is defined as

$$\mathrm{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$

A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more corrupted by noise.

Because SNR is the ratio of two powers, it is often described in decibel units, $\mathbf{SNR_{dB}}$, defined as

$$\mathrm{SNR_{dB}} = 10 \log_{10} \mathrm{SNR}$$

The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB}?

Solution

The values of SNR and SNR_{dB} can be calculated as follows:

$$\mathbf{SNR = (10{,}000\ \mu w) / (1\ \mu w) = 10{,}000 \qquad SNR_{dB} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40}$$

DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

- The bandwidth available
- The level of the signals we use
- The quality of the channel (the level of noise)

Two formulas were given to calculate the data rate:

- ✓ Nyquist for a noiseless channel
- ✓ Shannon for a noisy channel.

Noiseless Channel: Nyquist Bit Rate

For a noiseless channel, the **Nyquist bit rate** formula is

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

In this formula,

- bandwidth is the bandwidth of the channel
- L is the number of signal levels used to represent data
- BitRate is the bit rate in bits per second.

Increasing the levels of a signal may reduce the reliability of the system.

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L \longrightarrow \log_2 L = 6.625 \longrightarrow L = 2^{6.625} = 98.7 \text{ levels}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

Noisy Channel: Shannon Capacity

- In reality, we cannot have a noiseless channel; the channel is always noisy.
- In 1944, Claude Shannon introduced a formula, called the **Shannon capacity**

$$\textbf{Capacity} = \textbf{bandwidth} \times \log_2(1 + \textbf{SNR})$$

- ✓ bandwidth is the bandwidth of the channel
- ✓ SNR is the signal-to-noise ratio
- ✓ capacity is the capacity of the channel in bits per second.

- The formula defines a characteristic of the channel, not the method of transmission.

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2(1 + 3162) = 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

The signal-to-noise ratio is often given in decibels. Assume that $\text{SNR}_{\text{dB}} = 36$ and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$\text{SNR}_{\text{dB}} = 10 \log_{10}\text{SNR} \longrightarrow \text{SNR} = 10^{\text{SNR}_{\text{dB}}/10} \longrightarrow \text{SNR} = 10^{3.6} = 3981$$

$$C = B \log_2(1 + \text{SNR}) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2(1 + \text{SNR}) = 10^6 \log_2(1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \longrightarrow L = 4$$

The Shannon capacity gives us the upper limit;
the Nyquist formula tells us how many signal levels we need.

PERFORMANCE

Bandwidth

In networking, we use the term bandwidth in two contexts.

- ❑ The first, ***bandwidth in hertz***, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- ❑ The second, ***bandwidth in bits per second***, refers to the speed of bit transmission in a channel or link.

Throughput

The throughput is a measure of ***how fast*** we can actually send data through a network.

The **bandwidth** is a potential measurement of a link; the **throughput** is an actual measurement of how fast we can send data.

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution

We can calculate the throughput as

$$\textbf{Throughput} = (12{,}000 \times 10{,}000) / 60 = 2 \text{ Mbps}$$

The throughput is almost one-fifth of the bandwidth in this case.

Latency (Delay)

The **latency** or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

Latency = propagation time + transmission time + queuing time + processing delay

Propagation time = Distance / (Propagation Speed)

Transmission time = (Message size) / Bandwidth

Queuing Time

- It is the time needed for each intermediate or end device to hold the message before it can be processed.
- It is not a fixed factor; it changes with the load imposed on the network.
- When there is heavy traffic on the network, the queuing time increases.
- An intermediate device, such as a router, queues the arrived messages and processes them one by one. If there are many messages, each message will have to wait.

What are the propagation time and the transmission time for a 2.5-KB (kilobyte) message (an e-mail) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.

Solution

We can calculate the propagation and transmission time as

$$\textbf{Propagation time} = (12{,}000 \times 1000) / (2.4 \times 10^8) = 50 \text{ ms}$$

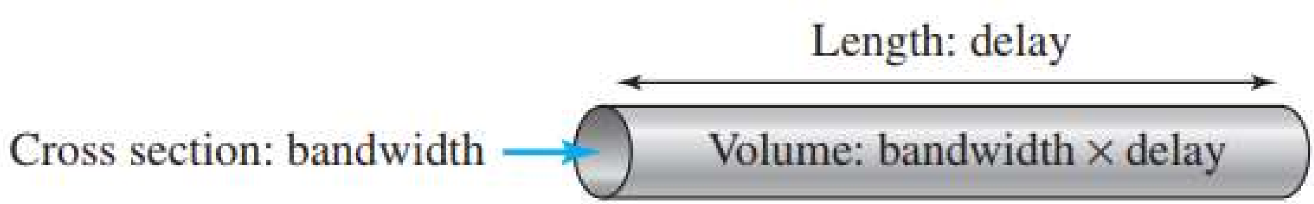
$$\textbf{Transmission time} = (2500 \times 8) / 10^9 = 0.020 \text{ ms}$$

Bandwidth-Delay Product

The bandwidth-delay product defines the number of bits that can fill the link.

We can think about the link between two points as a pipe. The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay. We can say the volume of the pipe defines the bandwidth-delay product

Concept of bandwidth-delay product



Jitter

Jitter is a problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example).

Digital Transmission

In digital transmission, if ***data are digital***, we need to use **digital-to-digital conversion techniques**.

If ***data are analog***, we need to use **analog-to-digital conversion**.

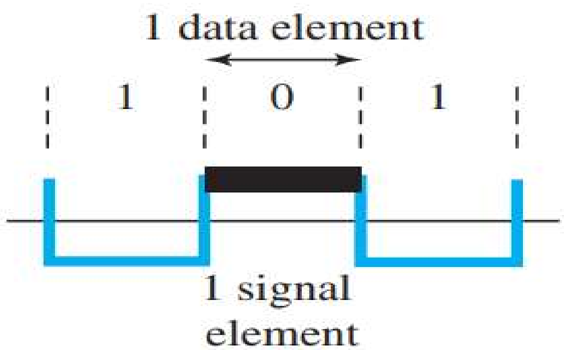
Digital-to-Digital Conversion

The conversion involves three techniques:

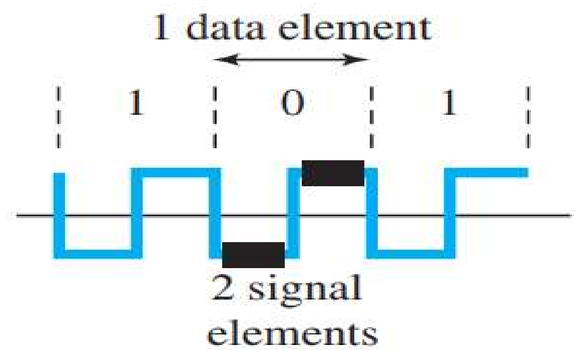
- ✓ Line coding
- ✓ Block coding
- ✓ Scrambling

Signal Element Versus Data Element

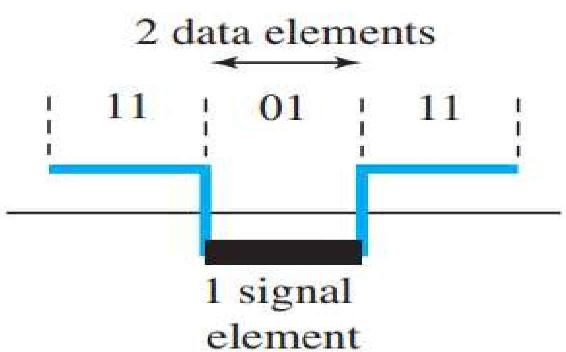
- A data element is the smallest entity that can represent a piece of information: this is the bit.
- In digital data communications, a signal element carries data elements.
- A signal element is the shortest unit (time-wise) of a digital signal.
- In other words, data elements are what we need to send; signal elements are what we can send.
- Data elements are being carried; signal elements are the carriers.



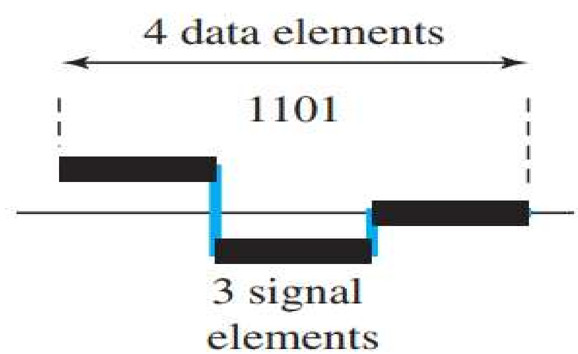
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements $\left(r = \frac{1}{2}\right)$



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements $\left(r = \frac{4}{3}\right)$

Data Rate Versus Signal Rate

- The data rate defines the number of data elements (bits) sent in 1s. The unit is **bits per second (bps)**.
- The signal rate is the number of signal elements sent in 1s. The unit is the **baud**.
- There are several common terminologies used. The ***data rate*** is sometimes called the ***bit rate***; the ***signal rate*** is sometimes called the ***pulse rate, the modulation rate***, or the ***baud rate***.
- Increasing the data rate increases the speed of transmission; decreasing the signal rate decreases the bandwidth requirement.

In general, the relationship between data rate (N) and signal rate (S)

$$S = N/r$$

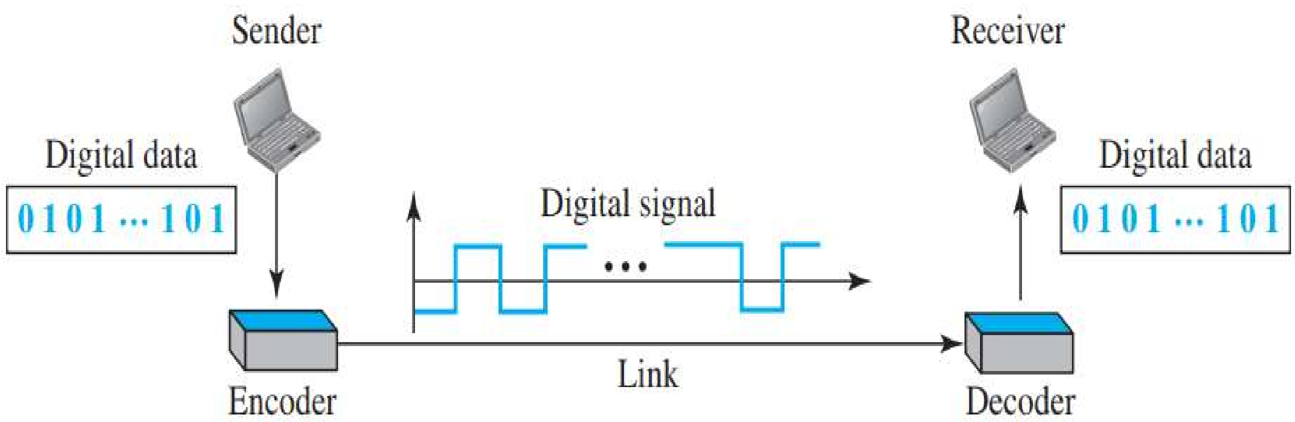
The relationship between data rate and signal rate as

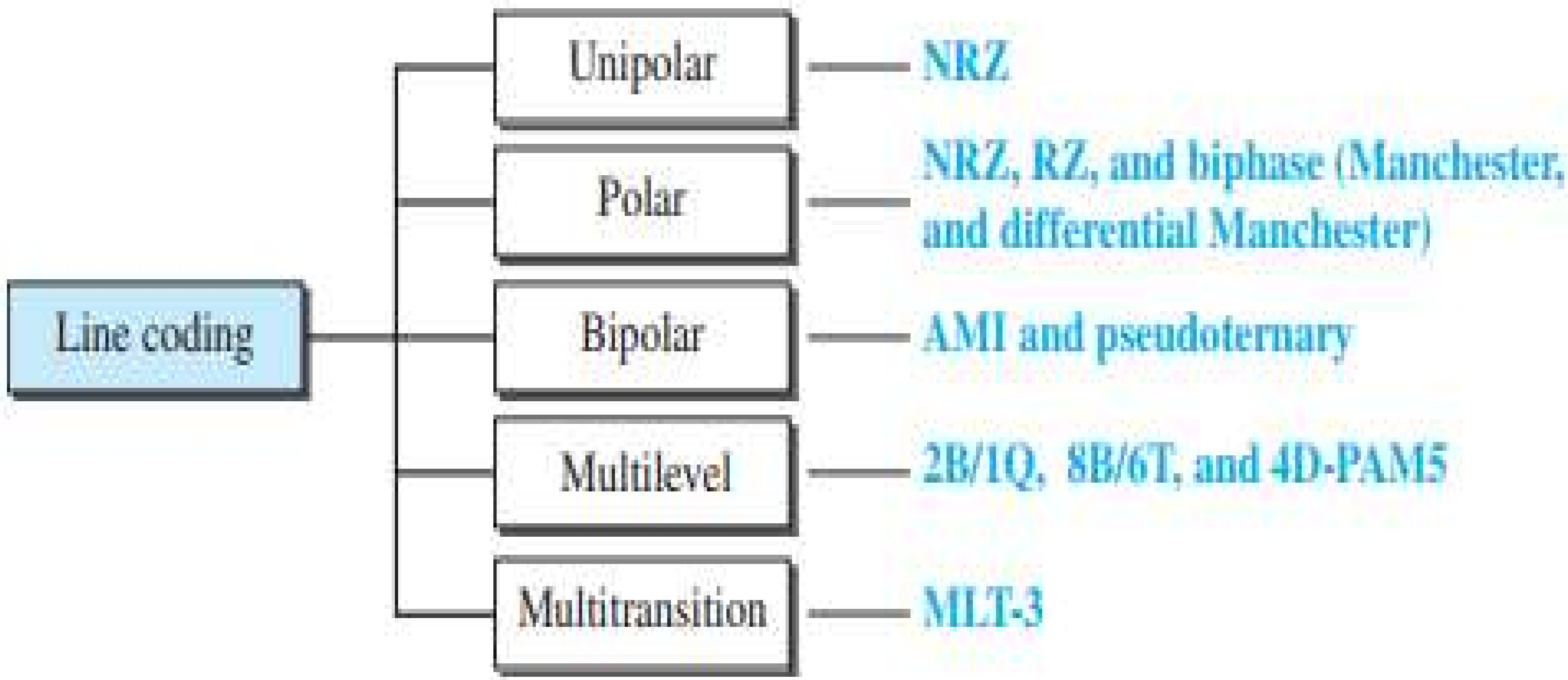
$$S_{ave} = c * N * (1/r) \text{ baud}$$

where N is the data rate (bps); c is the case factor, which varies for each case; S is the number of signal elements per second; and r is the previously defined factor.

Line Coding

- Line coding is the process of converting digital data to digital signals.
- We assume that data, in the form of text, numbers, graphical images, audio, or video, are stored in computer memory as sequences of bits.
- Line coding converts a sequence of bits to a digital signal.
- At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are re-created by decoding the digital signal.




Line coding
Unipolar
Polar
Bipolar
Multilevel
Multitransition
NRZ
NRZ, RZ, and biphase (Manchester, and differential Manchester)
AMI and pseudoternary
2B/1Q, 8B/6T, and 4D-PAM5
MLT-3

Nonreturn to Zero-Level (NRZ-L)

0 = high level

1 = low level

Nonreturn to Zero Inverted (NRZI)

0 = no transition at beginning of interval (one bit time)

1 = transition at beginning of interval

Bipolar-AMI

0 = no line signal

1 = positive or negative level, alternating for successive ones

Pseudoternary

0 = positive or negative level, alternating for successive zeros

1 = no line signal

Manchester

0 = transition from high to low in middle of interval

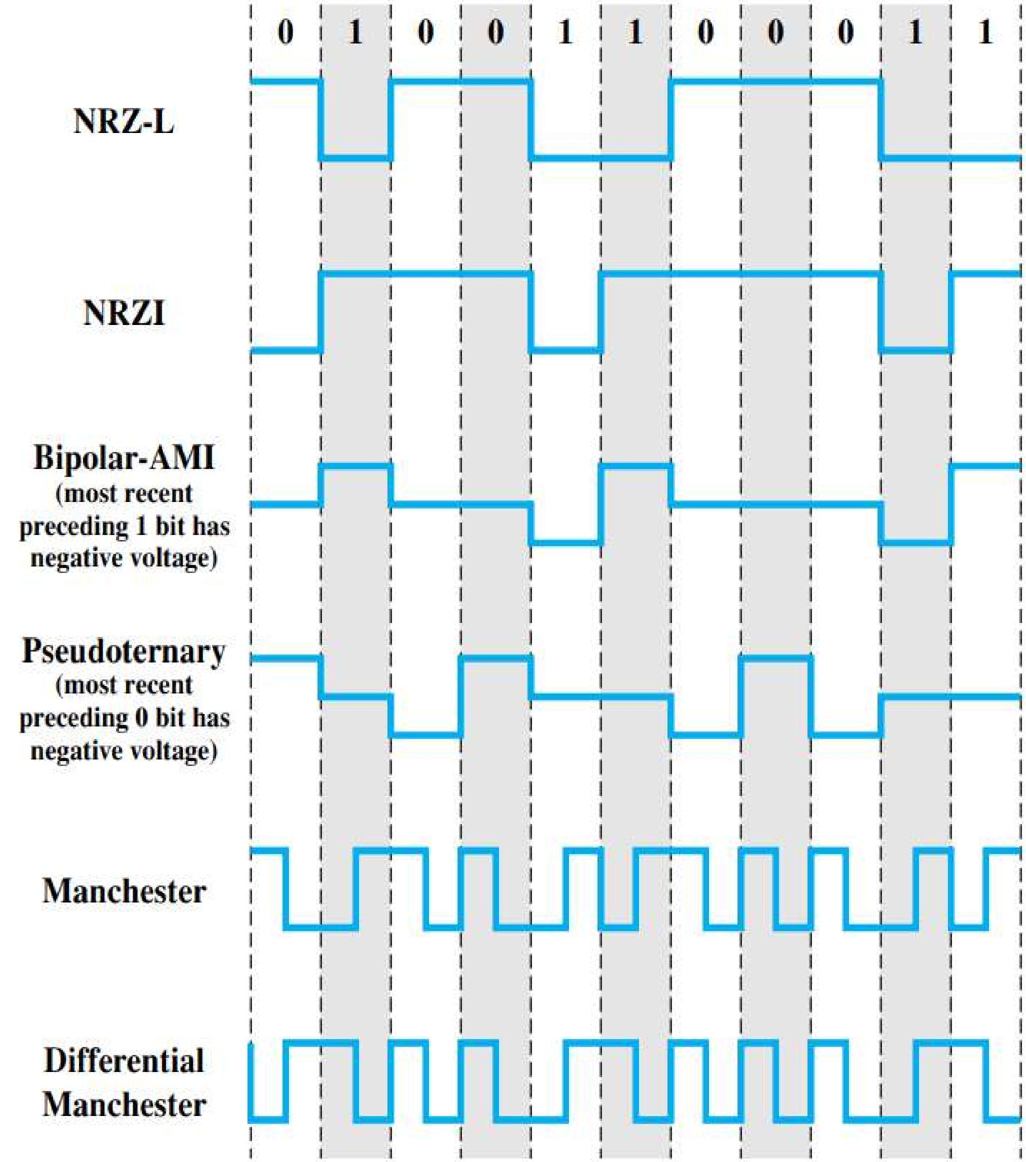
1 = transition from low to high in middle of interval

Differential Manchester

Always a transition in middle of interval

0 = transition at beginning of interval

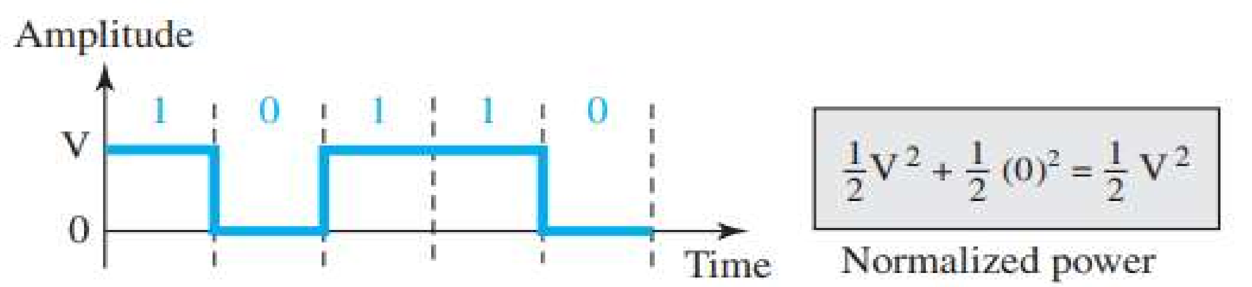
1 = no transition at beginning of interval



NRZ (Non-Return-to-Zero)

- A unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0.
- It is called NRZ because the signal does not return to zero at the middle of the bit.
- This scheme is very costly. The normalized power is double that for polar NRZ.

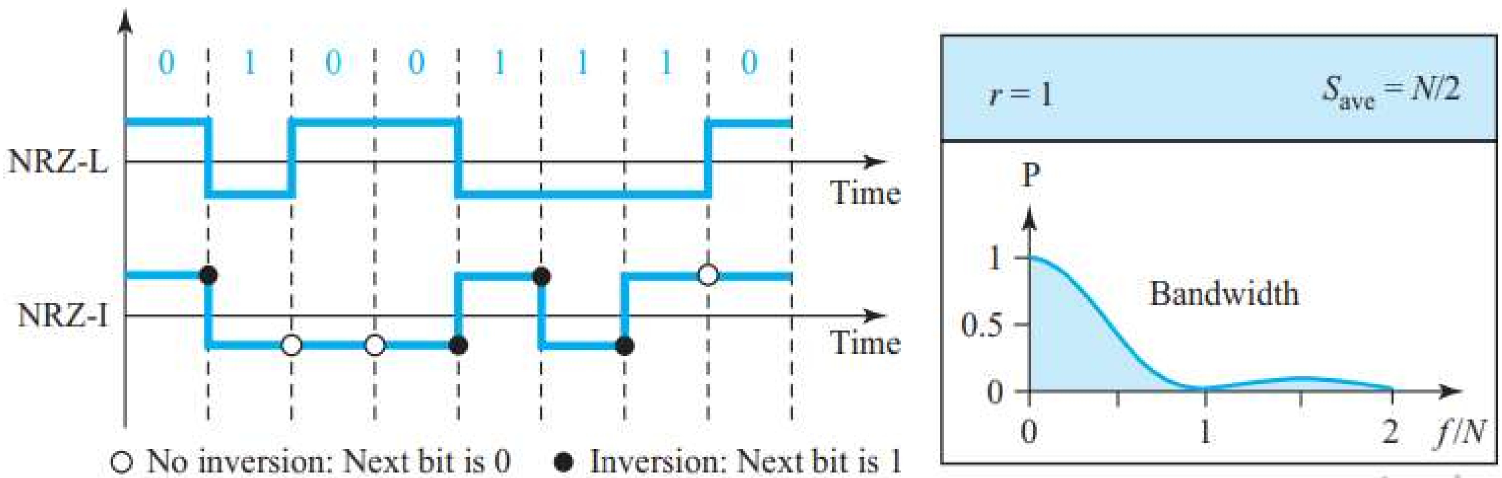
Unipolar NRZ scheme



** Normalized power → The power needed to send 1 bit per unit line resistance

Polar Schemes (NRZ-L and NRZ-I)

- In NRZ-L (NRZ-Level), the level of the voltage determines the value of the bit.
- In NRZ-I (NRZ-Invert), the change or lack of change in the level of the voltage determines the value of the bit. If there is no change, the bit is 0; if there is a change, the bit is 1.



- **Baseline wandering** is a problem for both variations
  - ✓ This problem is twice as severe in NRZ-L. If there is a long sequence of 0s or 1s in NRZ-L, the average signal power becomes skewed. The receiver might have difficulty discerning the bit value.
  - ✓ In NRZ-I this problem occurs only for a long sequence of 0s.
  - ✓ Solution: Eliminating the long sequence of 0s can avoid baseline wandering.
- NRZ-L and NRZ-I both have a **DC component problem**.
- **Synchronization problem** (sender and receiver clocks are not synchronized) also exists in both schemes.
  - ✓ This problem is more serious in NRZ-L than in NRZ-I. While a long sequence of 0s can cause a problem in both schemes, a long sequence of 1s affects only NRZ-L.
- Another problem with NRZ-L occurs when there is a **sudden change of polarity** in the system. For example, if twisted-pair cable is the medium, a change in the polarity of the wire results in all 0s interpreted as 1s and all 1s interpreted as 0s.
  - ✓ **NRZ-I does not have this problem.**

NRZ-L and NRZ-I both have an average signal rate of $N/2$ Bd.

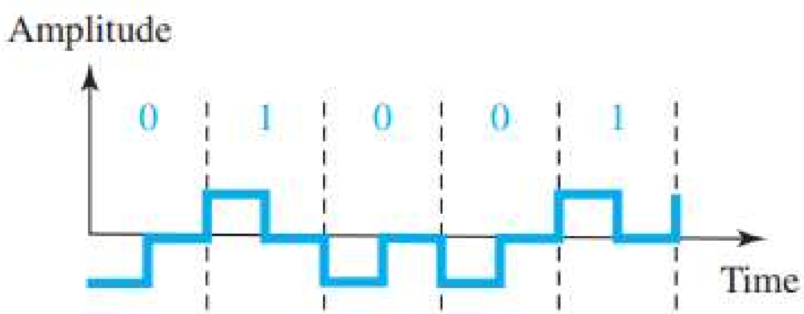
A system is using NRZ-I to transfer 10-Mbps data. What are the average signal rate and minimum bandwidth?

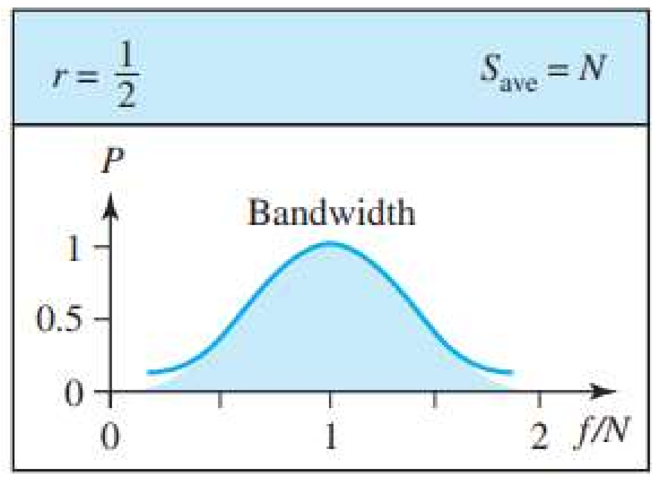
Solution

The average signal rate is $S = N/2 = 500$ kbaud. The minimum bandwidth for this average baud rate is $B_{\min} = S = 500$ kHz.

Return-to-Zero (RZ)

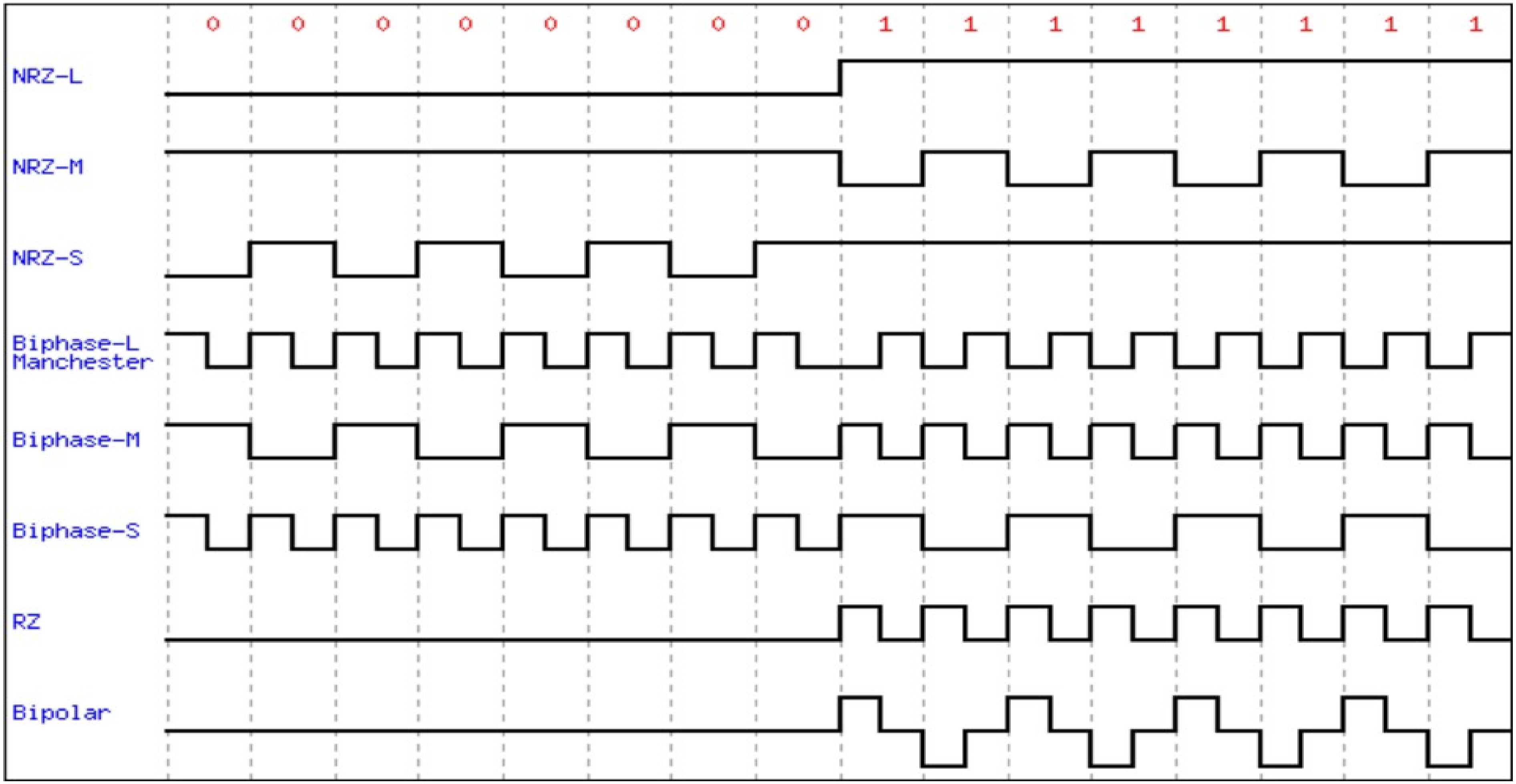
- Synchronization problem in NRZ.
- One **solution** is the Polar Return-to-Zero (RZ) or simply RZ scheme, which uses three values: positive, negative, and zero. In RZ, the signal changes not between bits but during the bit.
- The signal goes to 0 in the middle of each bit. It remains there until the beginning of the next bit.



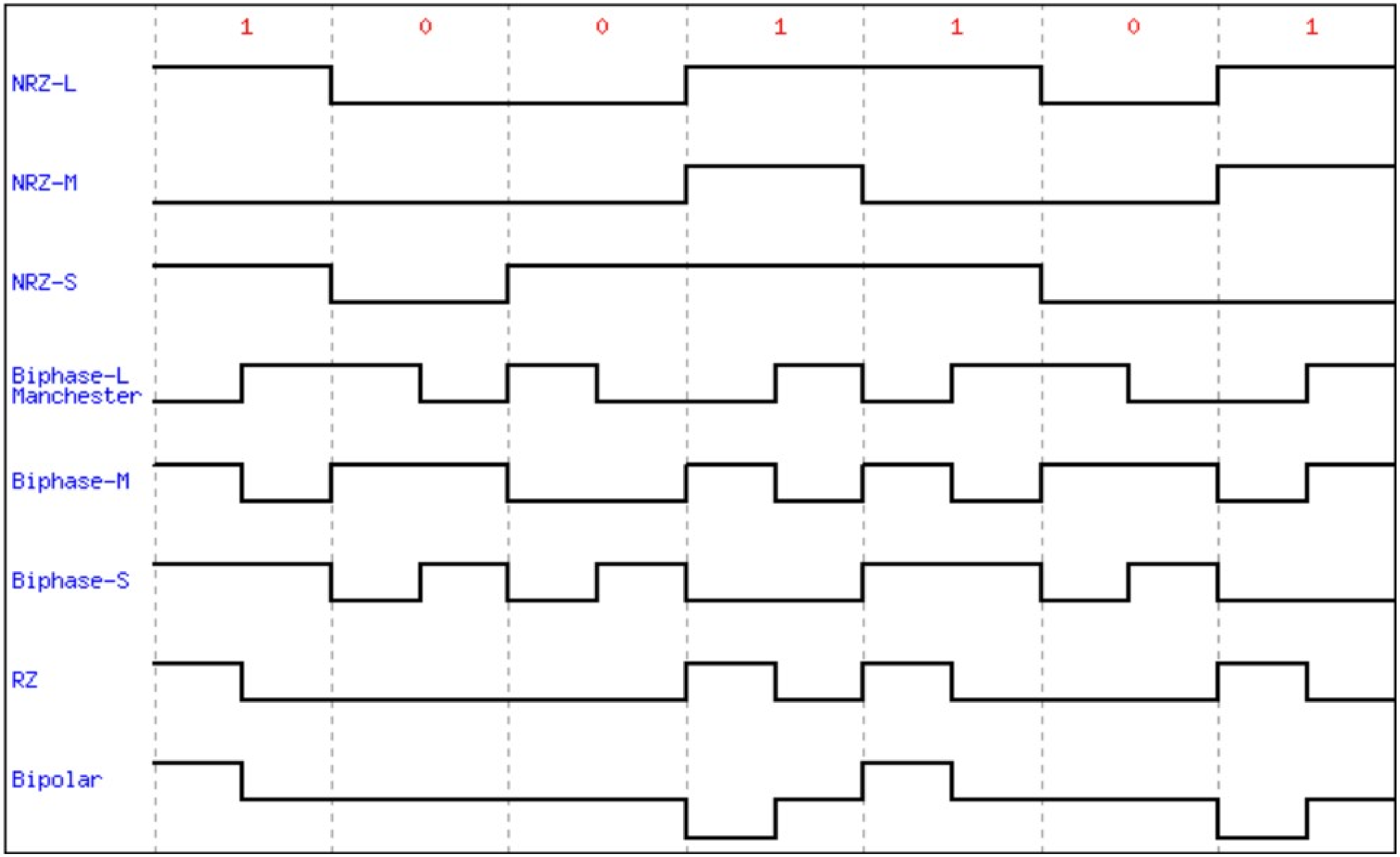


- The main **disadvantage** of RZ encoding is that it requires two signal changes to encode a bit and therefore **occupies greater bandwidth**.
- The **problem**, a **sudden change of polarity** resulting in all 0s interpreted as 1s and all 1s interpreted as 0s, still exists here.
- Another **problem** is the **complexity**: RZ uses three levels of voltage, which is more complex to create and discern.
- In this scheme, there is **no DC component problem**.

Digital Data or Binary value : 0000000011111111



Digital data: 1001101



Block Coding

- We need redundancy to ensure synchronization and to provide some kind of inherent error detecting.
- Block coding can give us this redundancy and improve the performance of line coding.
- In general, block coding ***changes a block of m bits into a block of n bits***, where n is larger than m.
- Block coding is referred to as an **mB/nB** encoding technique.

Block coding normally involves three steps: *division*, *substitution*, and *combination*.

- ✓ In the **division step**, a sequence of bits is divided into groups of m bits. For example, in 4B/5B encoding, the original bit sequence is divided into 4-bit groups.
- ✓ The heart of block coding is the **substitution** step. In this step, we substitute an m-bit group for an n-bit group. For example, in the case of 4B/5B encoding, we substitute a 4-bit code for a 5-bit group.
- ✓ Finally, the n-bit groups are **combined** to form a stream.

Scrambling Techniques

- Sequences that would result in a constant voltage level on the line are replaced by filling sequences that will provide sufficient transitions for the receiver’s clock to maintain synchronization.
- The filling sequence must be recognized by the receiver and replaced with the original data sequence.
- The filling sequence is the same length as the original sequence, so there is no data rate penalty

Bipolar with 8-Zeros Substitution (B8ZS) commonly used in North America

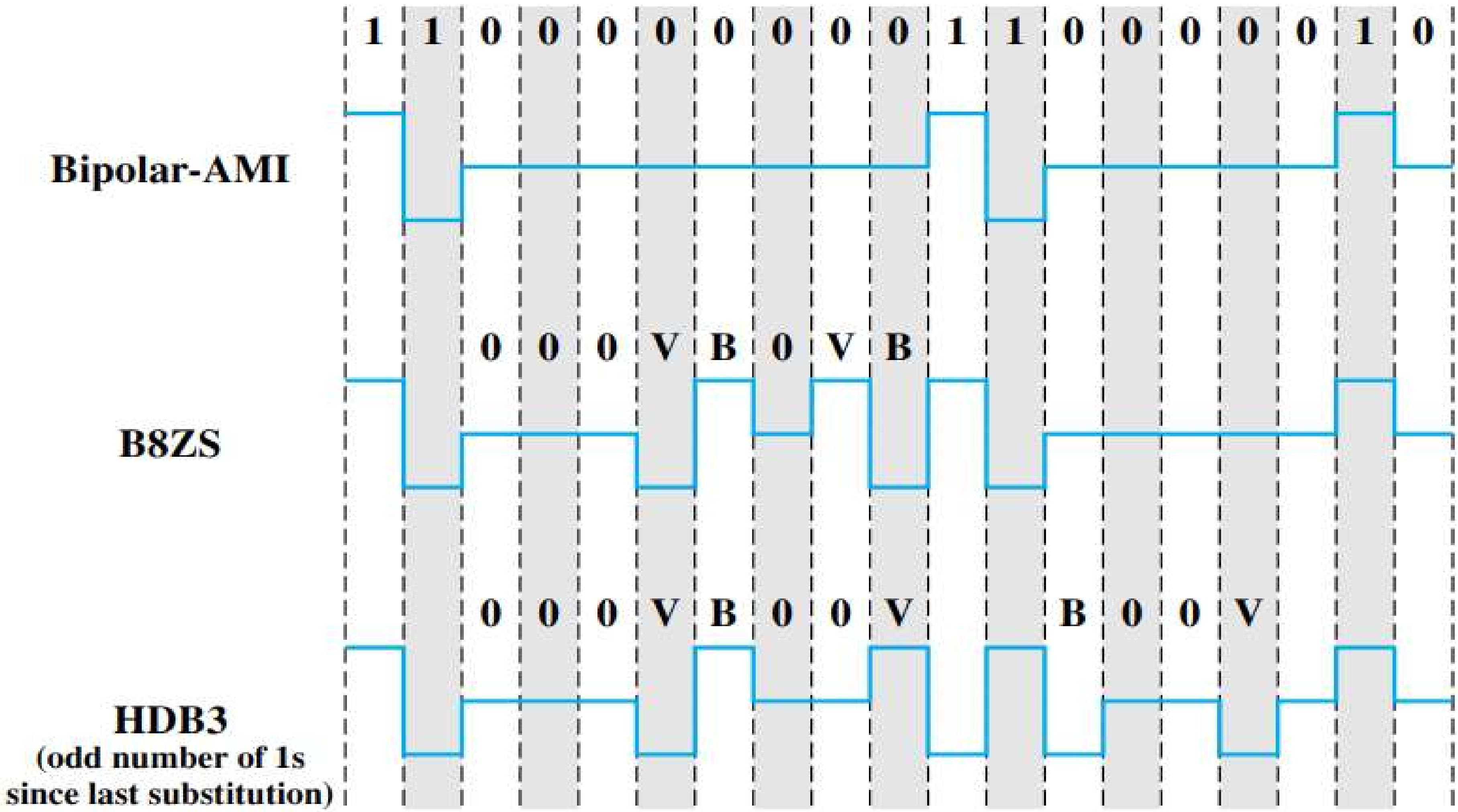
To overcome the drawback of the AMI code, where a long string of zeros may result in loss of synchronization, the encoding is amended with the following rules:

- If an octet of all zeros occurs and the last voltage pulse preceding this octet was positive, then the eight zeros of the octet are encoded as 000+−0−+.
- If an octet of all zeros occurs and the last voltage pulse preceding this octet was negative, then the eight zeros of the octet are encoded as 000−+0+−.

High-Density Bipolar-3 Zeros (HDB3) code commonly used in Europe and Japan

HDB3 Substitution Rules

	Number of Bipolar Pulses (ones) since Last Substitution	
Polarity of Preceding Pulse	**Odd**	**Even**
−	000−	+00+
+	000+	−00−



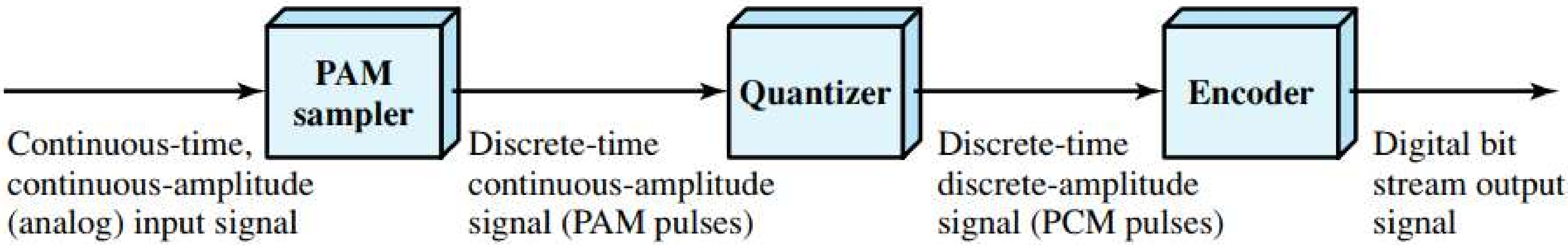
B = Valid bipolar signal
V = Bipolar violation

Analog-to-Digital Conversion

- The tendency today is to change an analog signal to digital data because the digital signal is less susceptible to noise.
- Two techniques, **pulse code modulation** and **delta modulation** are used here.
- After the digital data are created, one of the techniques in Digital-to-Digital is used to convert the digital data to a digital signal.

Pulse Code Modulation (PCM)

- The most common technique used to change an analog signal to digital data (digitization) is called pulse code modulation (PCM).
- A PCM encoder has three processes – Sampling, Encoding, and Quantizing



The three processes are:

1. The analog signal is sampled every T s.

2. The sampled signal is quantized, which means every sample is considered as a pulse.

3. The quantized values (pulses) are encoded as streams of bits.

PCM Bandwidth

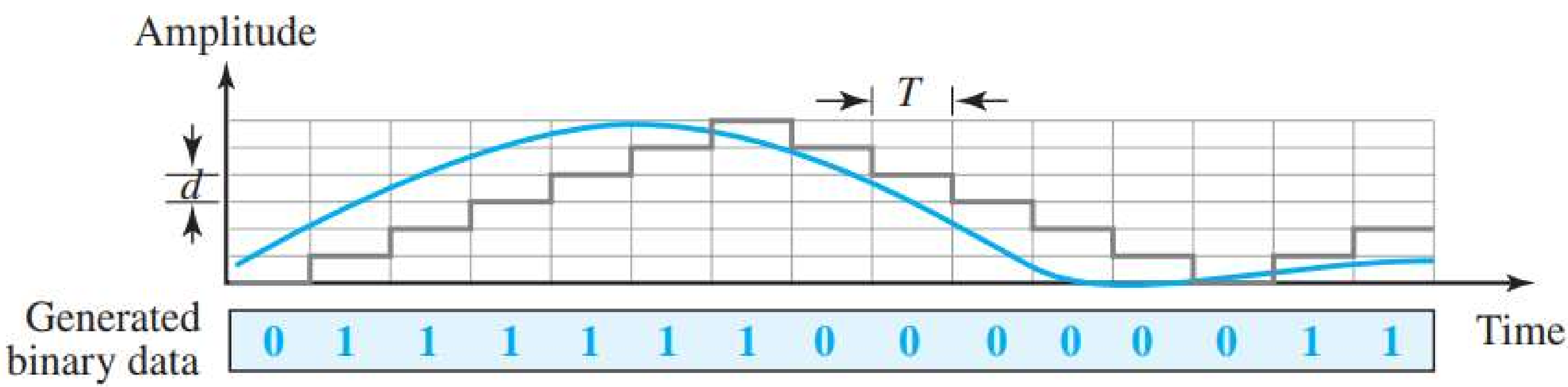
- It can be proved that the minimum bandwidth of the digital signal is

$$\mathbf{B_{min} = n_b \times B_{analog}}$$

- This means the minimum bandwidth of the digital signal is n_b times greater than the bandwidth of the analog signal.

Delta Modulation (DM)

PCM is a very complex technique. Other techniques have been developed to reduce the complexity of PCM. The simplest is delta modulation. PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample. There are no code words in this process; bits are sent one after another.

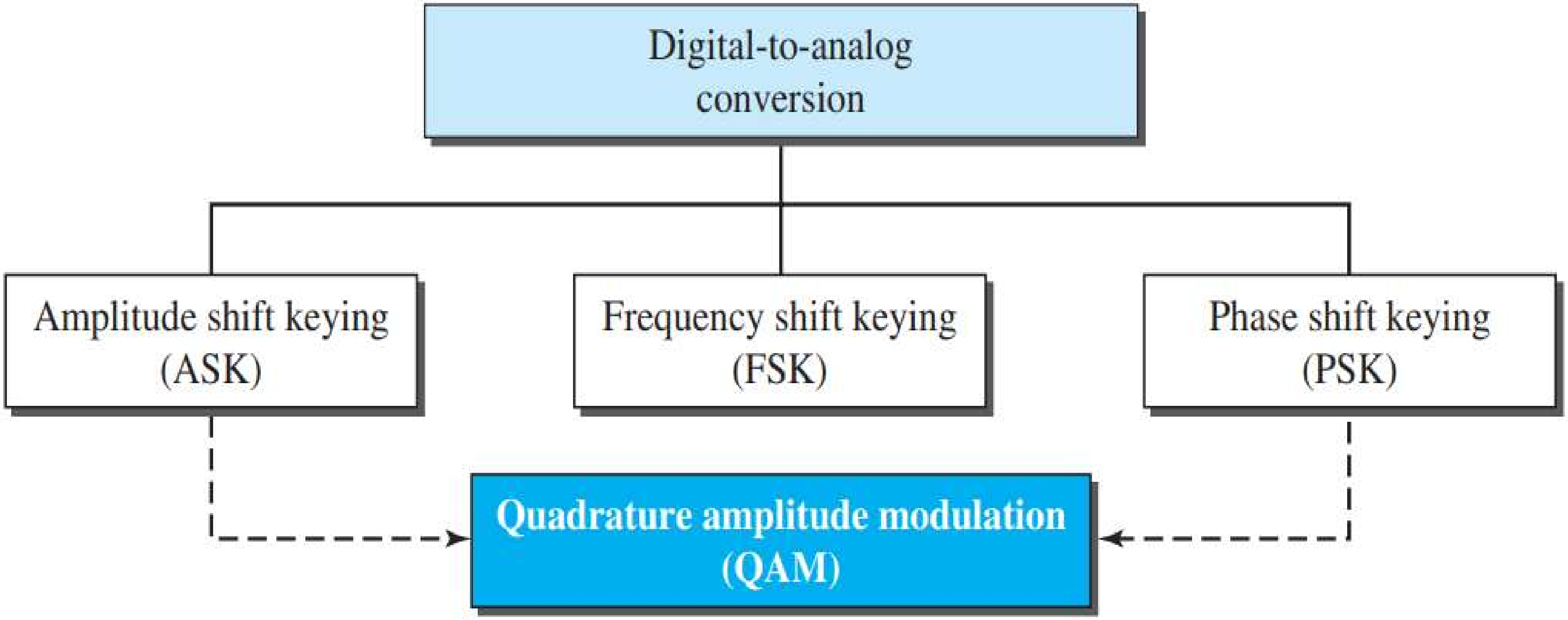


Analog Transmission

- Although digital transmission is desirable, it needs a low-pass channel (a channel that starts from 0); analog transmission is the only choice if we have a bandpass channel (a channel that does not start from zero).
- Converting **digital data to a bandpass analog signal** is traditionally called ***digital-to-analog conversion***.
- Converting a **low-pass analog signal to a bandpass analog signal** is traditionally called ***analog-to-analog conversion.***

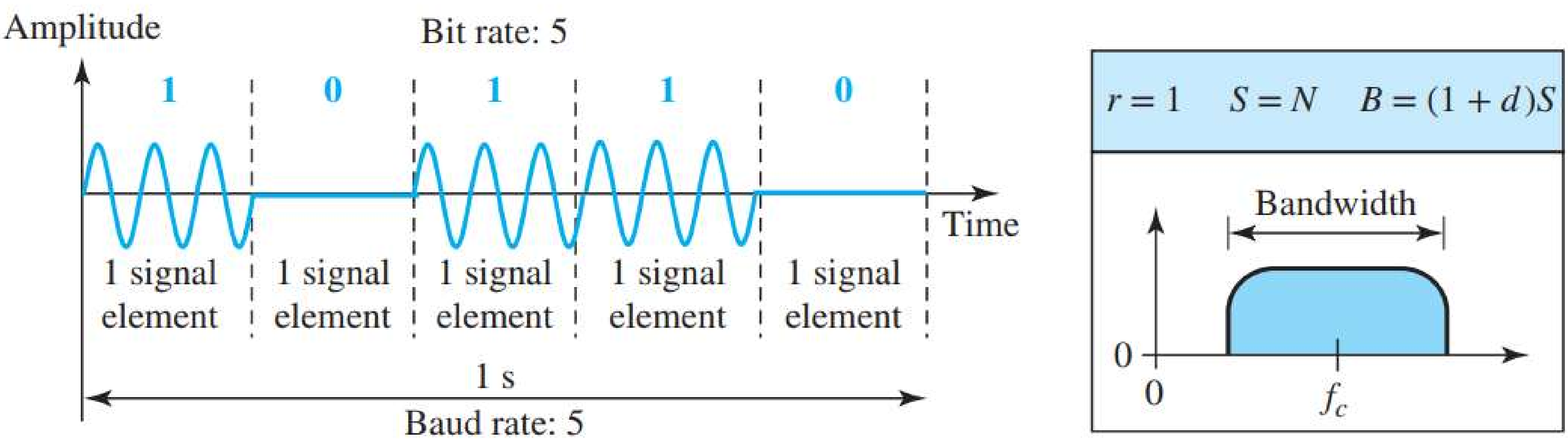
Digital-to-Analog Conversion

- Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.
- A sine wave is defined by three characteristics: **amplitude**, **frequency**, and **phase**.
- Any of the three characteristics can be altered in this way, giving us at least three mechanisms for *modulating* digital data into an analog signal:
    - Amplitude shift keying (ASK)
    - Frequency shift keying (FSK)
    - Phase shift keying (PSK).
- In addition, there is a fourth (and better) mechanism that combines changing both the amplitude and phase, called Quadrature amplitude modulation (QAM). QAM is the most efficient of these options and is the mechanism commonly used today.


Digital-to-analog conversion
Amplitude shift keying (ASK)
Frequency shift keying (FSK)
Phase shift keying (PSK)
Quadrature amplitude modulation (QAM)

Amplitude Shift Keying

In amplitude shift keying, the amplitude of the carrier signal is varied to create signal elements. Both frequency and phase remain constant while the amplitude changes.



We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with $d = 1$?

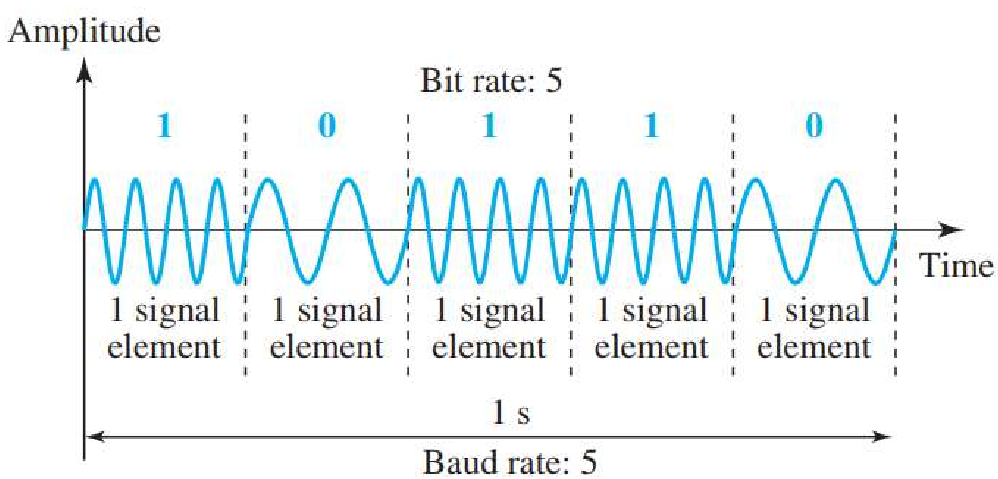
Solution

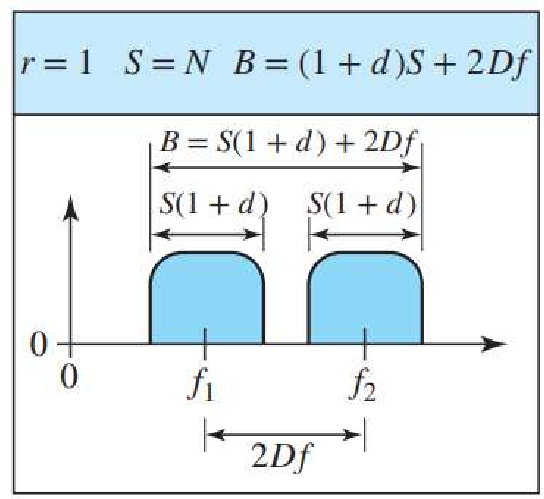
The middle of the bandwidth is located at 250 kHz. This means that our carrier frequency can be at $f_c = 250$ kHz. We can use the formula for bandwidth to find the bit rate (with $d = 1$ and $r = 1$).

$$B = (1 + d) \times S = 2 \times N \times (1/r) = 2 \times N = 100 \text{ kHz} \longrightarrow N = 50 \text{ kbps}$$

Frequency Shift Keying (FSK)

- In frequency shift keying, the frequency of the carrier signal is varied to represent data.
- The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes. Both the peak amplitude and phase remain constant for all signal elements.





We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with $d = 1$?

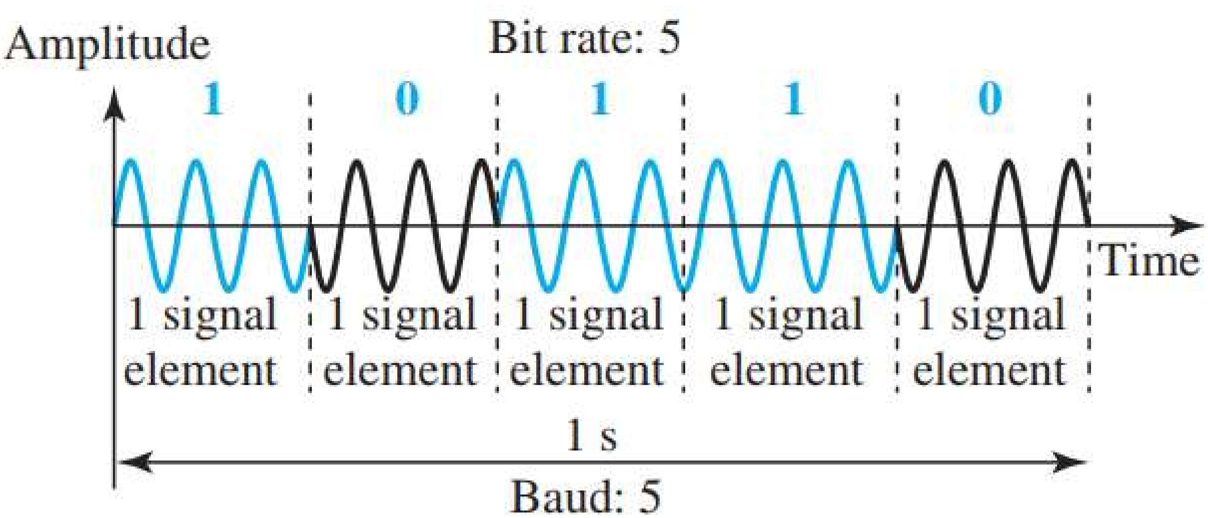
Solution

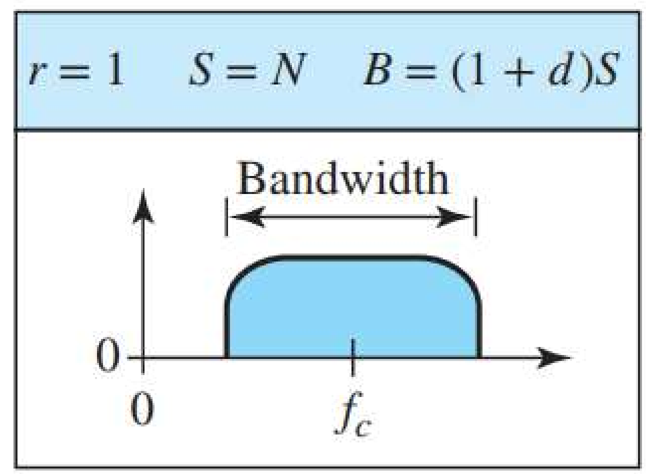
Contrary to above the example we are modulating by using FSK. The midpoint of the band is at 250 kHz. We choose $2\Delta_f$ to be 50 kHz; this means

$$B = (1 + d) \times S + 2\Delta_f = 100 \longrightarrow 2S = 50\text{ kHz} \longrightarrow S = 25\text{ kbaud} \longrightarrow N = 25\text{ kbps}$$

Phase Shift Keying

In phase shift keying, the phase of the carrier is varied to represent two or more different signal elements. Both peak amplitude and frequency remain constant as the phase changes.





Analog-to-Analog Conversion

Analog-to-analog conversion can be accomplished in three ways:

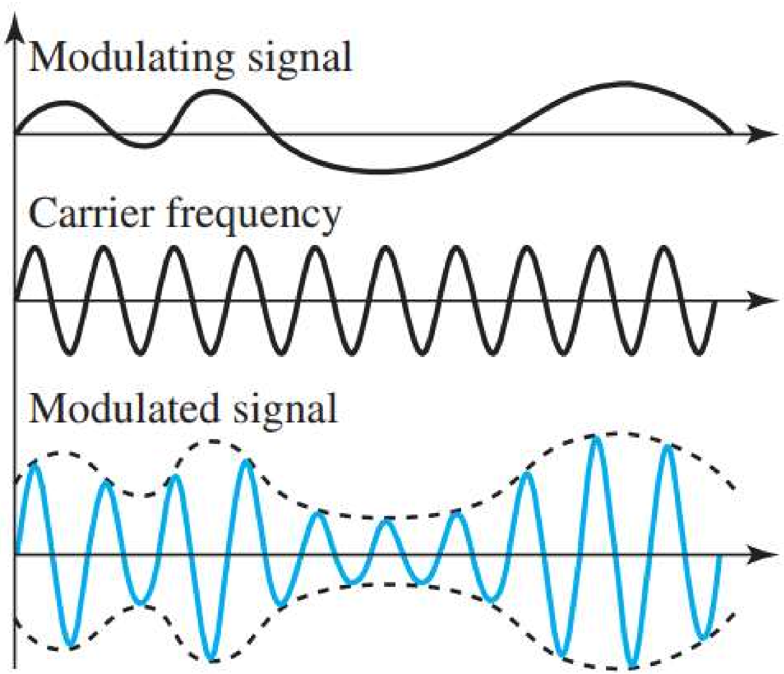
- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

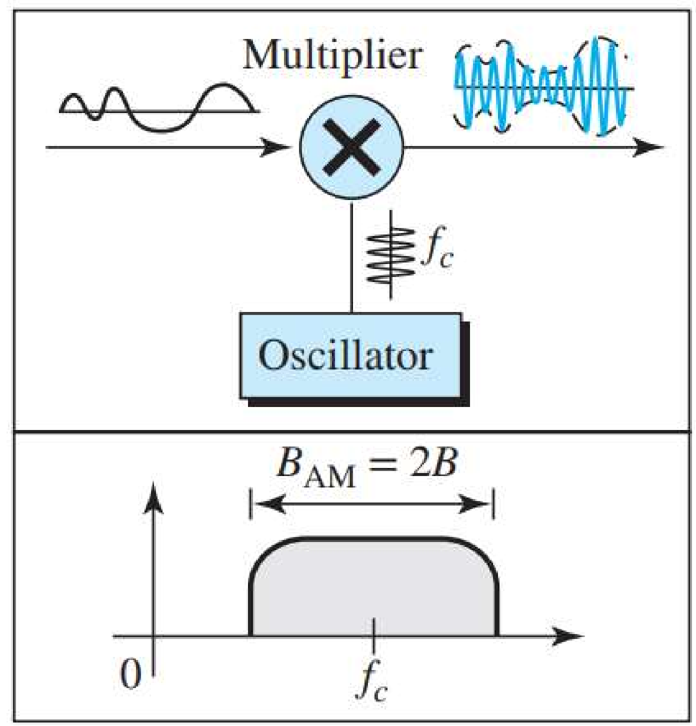
FM and PM are usually categorized together

Amplitude Modulation

- In AM transmission, the carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal.
- The frequency and phase of the carrier remain the same; only the amplitude changes to follow variations in the information.
- AM is normally implemented by using a simple multiplier because the amplitude of the carrier signal needs to be changed according to the amplitude of the modulating signal.
- The modulation creates a bandwidth that is twice the bandwidth of the modulating signal and covers a range centered on the carrier frequency.

- However, the signal components above and below the carrier frequency carry exactly the same information. For this reason, some implementations discard one-half of the signals and cut the bandwidth in half.





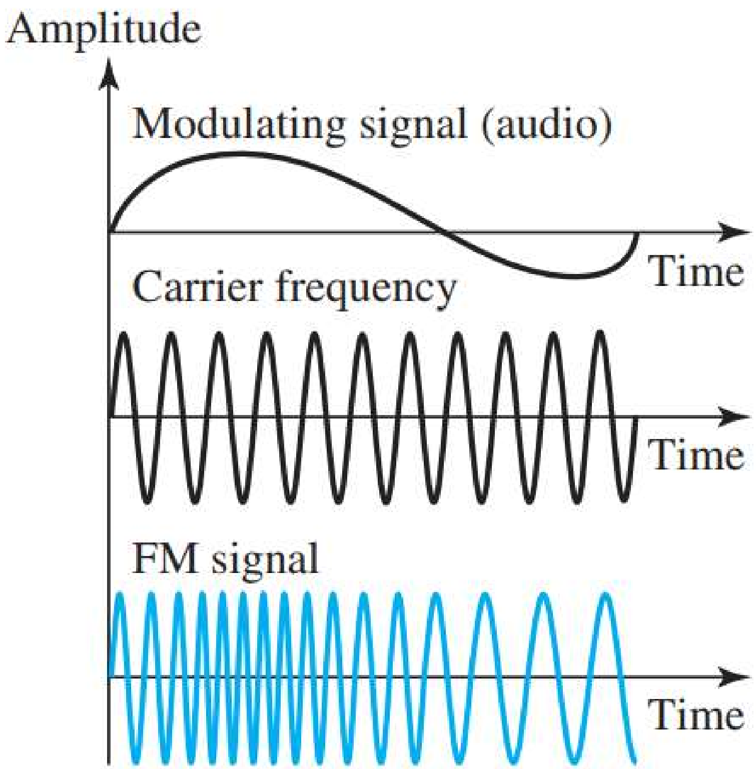
Frequency Modulation

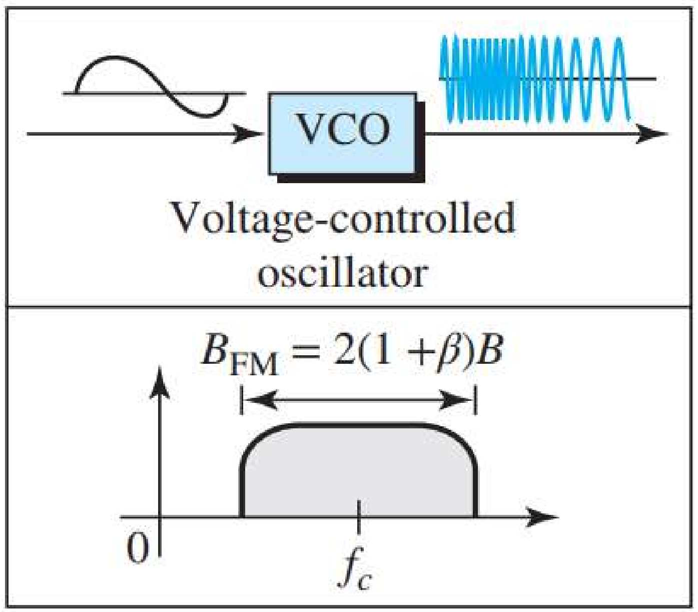
- In FM transmission, the frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal.
- The peak amplitude and phase of the carrier signal remain constant, but as the amplitude of the information signal changes, the frequency of the carrier changes correspondingly.
- FM is normally implemented by using a voltage-controlled oscillator.
- The frequency of the oscillator changes according to the input voltage, which is the amplitude of the modulating signal.

- The actual bandwidth is difficult to determine exactly, but it can be shown empirically as

$$B_{FM} = 2(1 + \beta)B$$

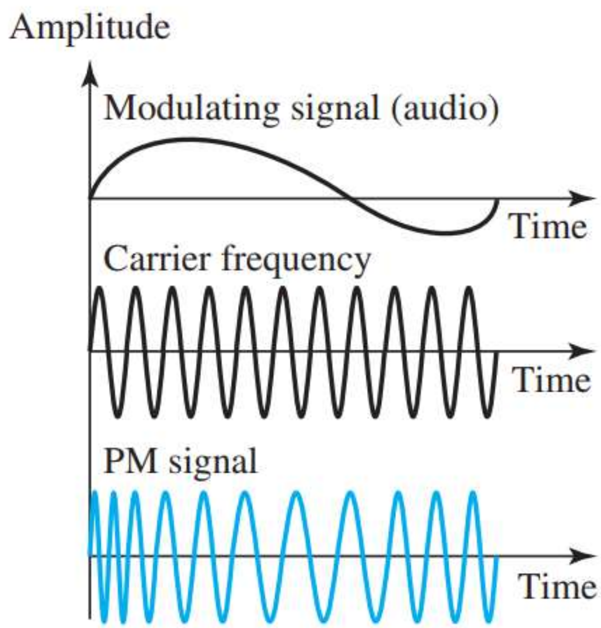
where β is a factor that depends on modulation technique with a common value of 4.

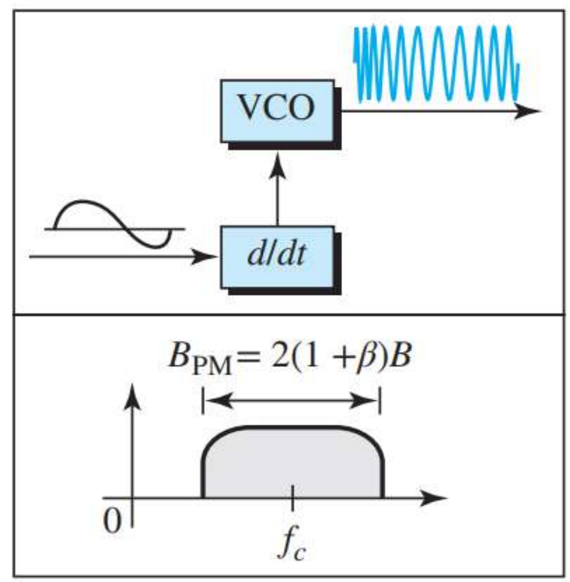




Phase Modulation

- In PM transmission, the phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal.
- The peak amplitude and frequency of the carrier signal remain constant, but as the amplitude of the information signal changes, the phase of the carrier changes correspondingly.

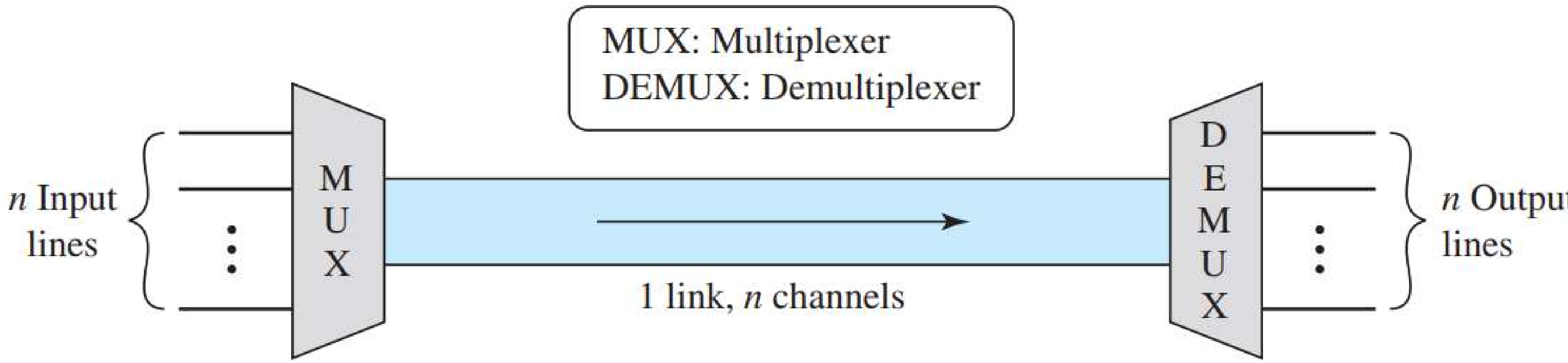




MULTIPLEXING

- Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.
- Today's technology includes high-bandwidth media such as optical fiber and terrestrial and satellite microwaves.
- Each has a bandwidth far in excess of that needed for the average transmission signal.
- If the bandwidth of a link is greater than the bandwidth needs of the devices connected to it, the bandwidth is wasted.
- An efficient system maximizes the utilization of all resources; bandwidth is one of the most precious resources we have in data communications.

In a multiplexed system, n lines share the bandwidth of one link.



- The word **link** refers to the physical path.
- The word **channel** refers to the portion of a link that carries a transmission between a given pair of lines.
    - One link can have many (n) channels.

There are three basic multiplexing techniques:

- frequency-division multiplexing (FDM),
- wavelength-division multiplexing (WDM)
- time-division multiplexing (TDM).

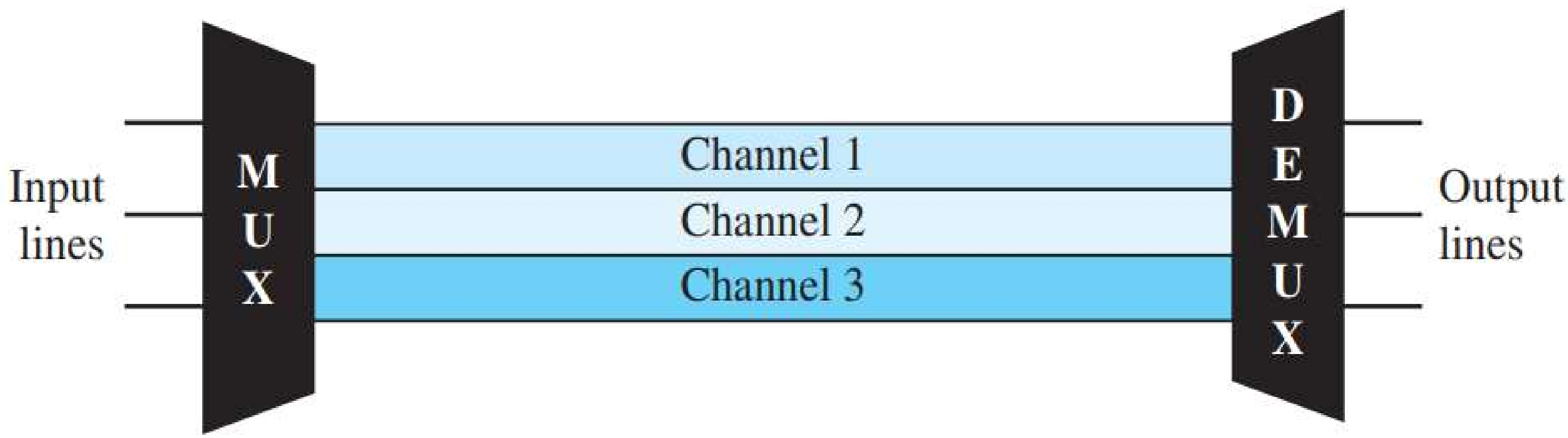
For Analog signals

For Digital signals

Frequency-Division Multiplexing

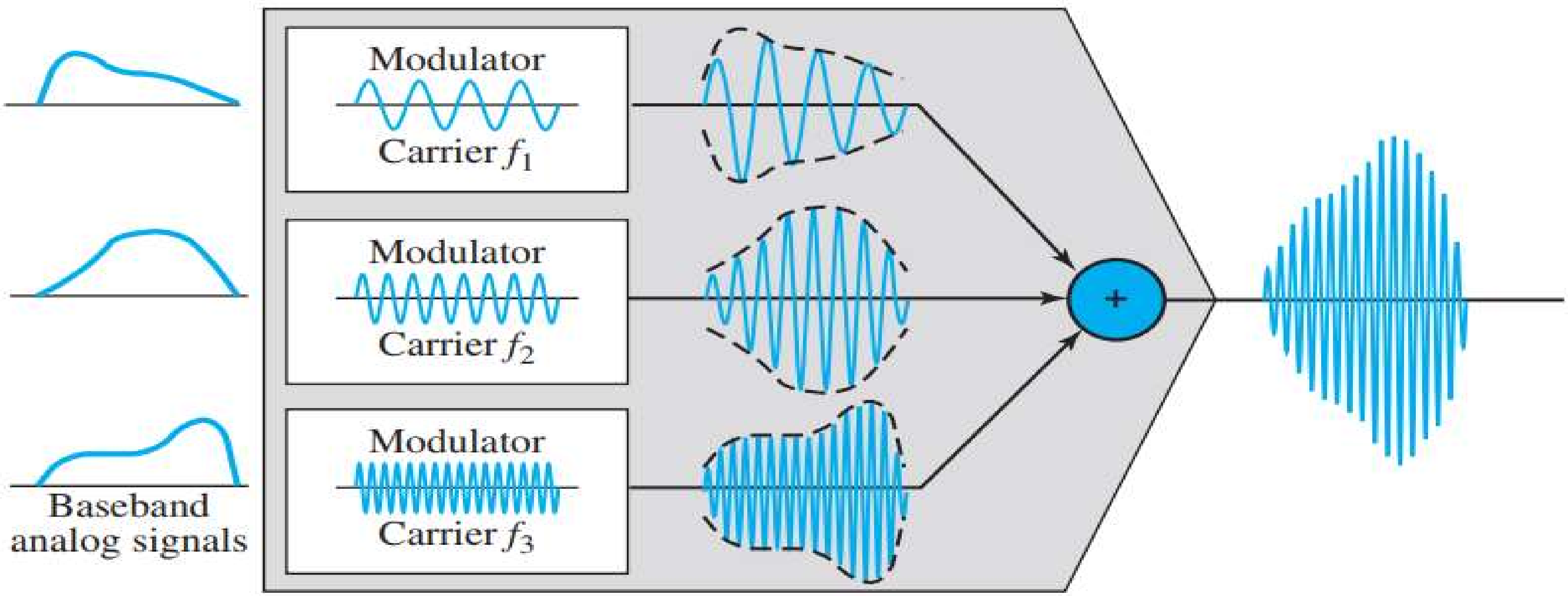
- Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.
- In FDM, signals generated by each sending device modulate different carrier frequencies.
- These modulated signals are then combined into a single composite signal that can be transported by the link.
- Carrier frequencies are separated by sufficient bandwidth to accommodate the modulated signal.

- These bandwidth ranges can be thought of as channels through which the various signals travel.
- Channels can be separated by strips of unused bandwidth— **guard bands** —to prevent signals from overlapping.
- In addition, carrier frequencies must not interfere with the original data frequencies.



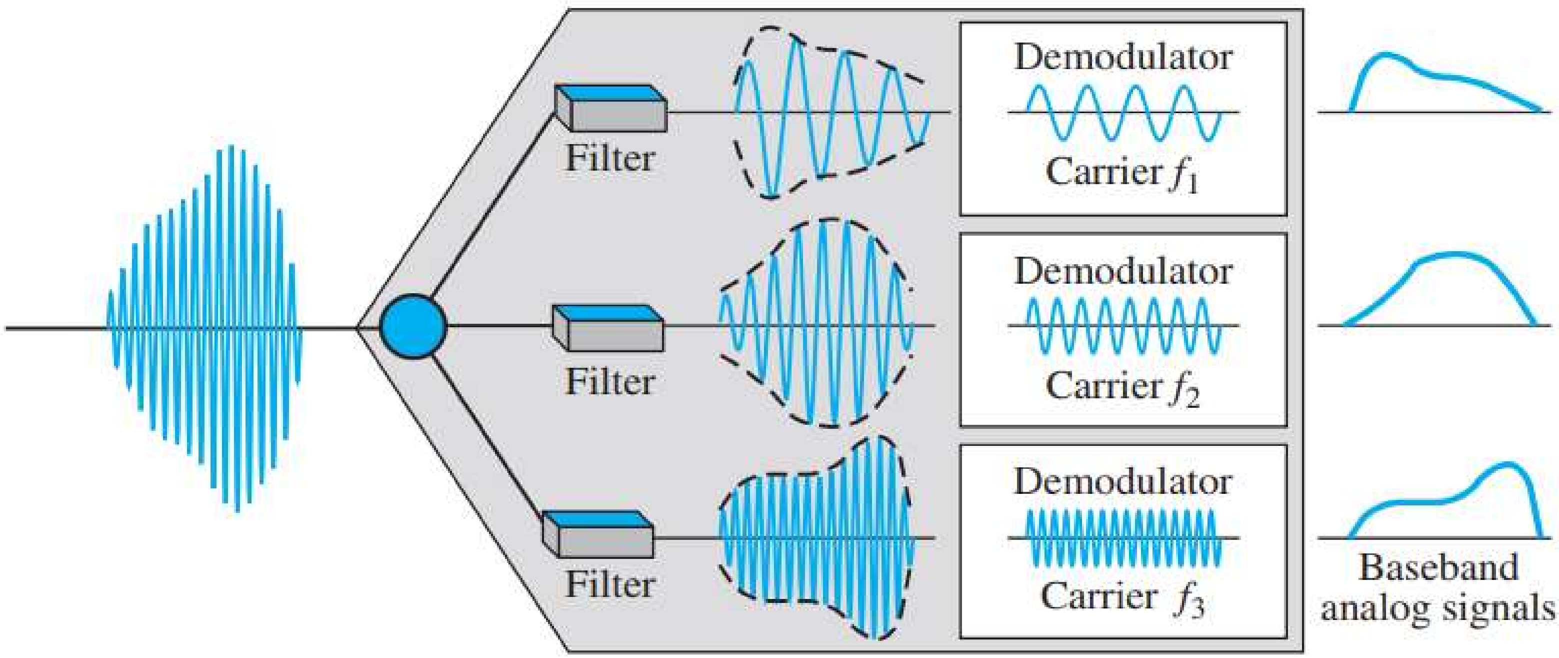
Multiplexing Process

- Each source generates a signal of a similar frequency range.
- Inside the multiplexer, these similar signals modulate different carrier frequencies (f1, f2, and f3).
- The resulting modulated signals are then combined into a single composite signal that is sent out over a media link that has enough bandwidth to accommodate it.



Demultiplexing Process

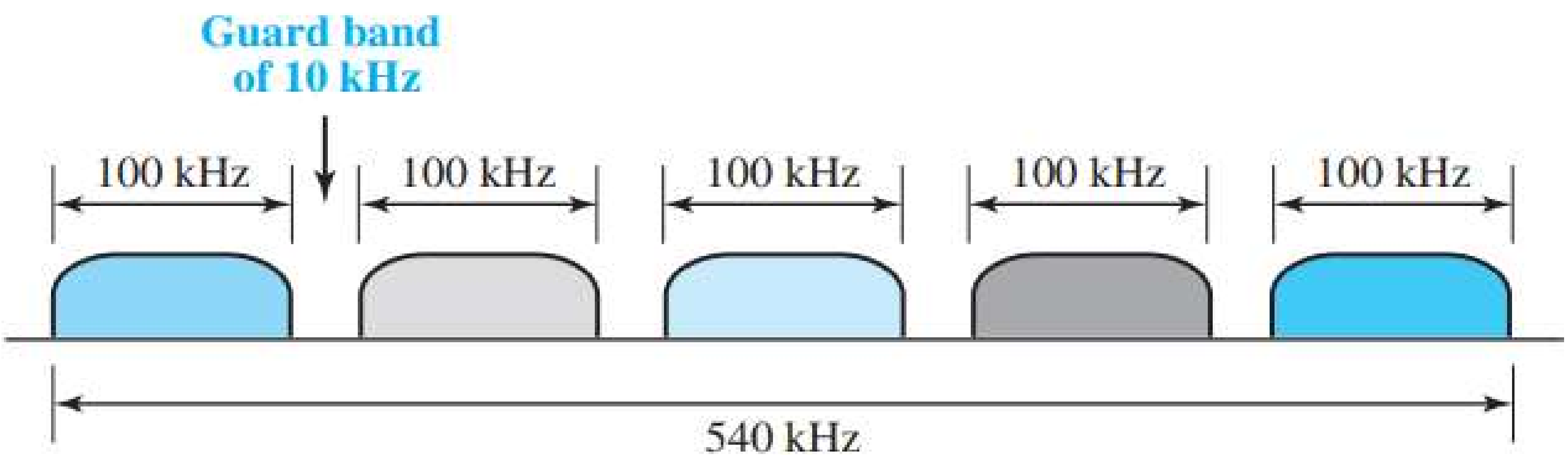
- The demultiplexer uses a series of filters to decompose the multiplexed signal into its constituent component signals.
- The individual signals are then passed to a demodulator that separates them from their carriers and passes them to the output lines.



Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

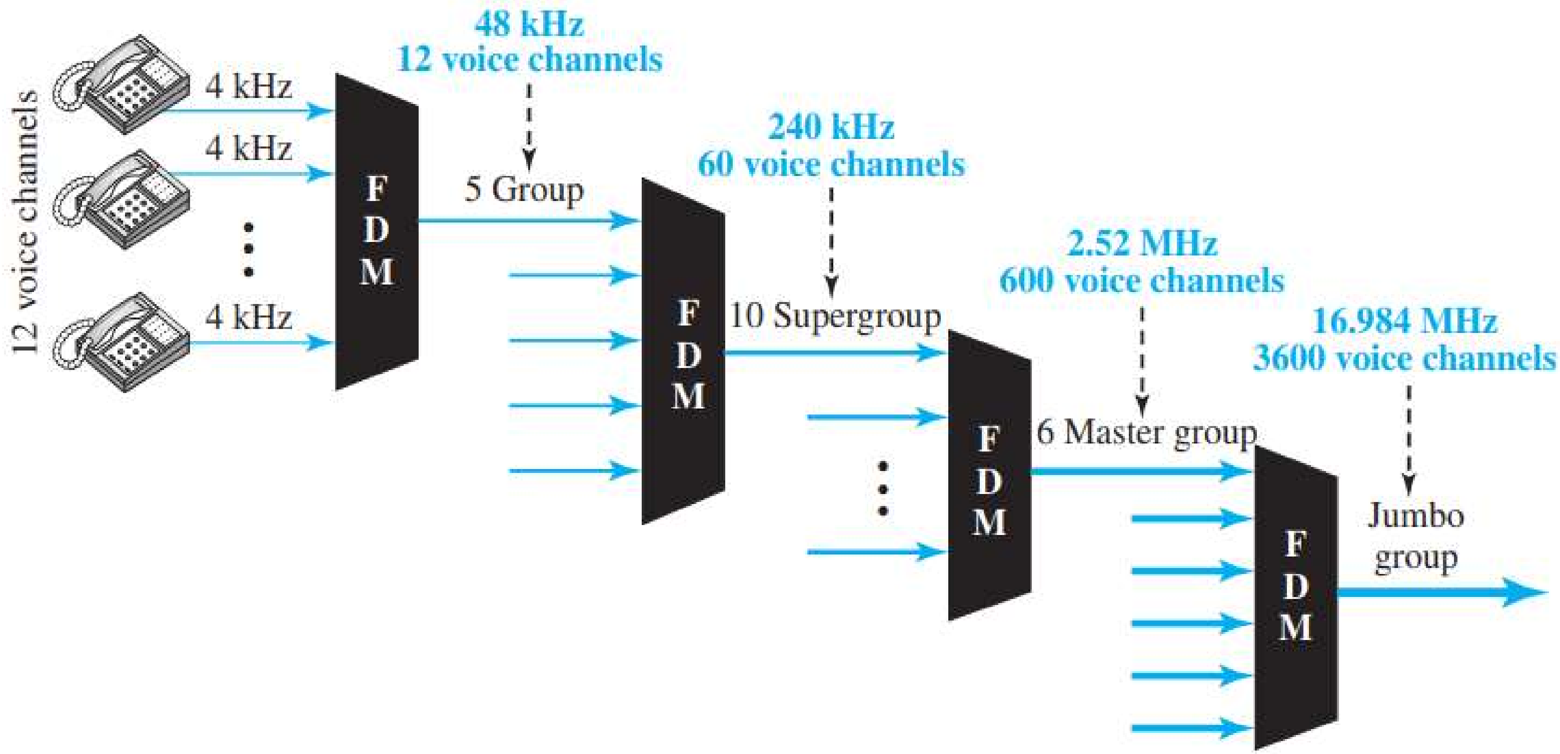
Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is at least $5 \times 100 + 4 \times 10 = 540$ kHz



Analog Carrier System

- To maximize the efficiency of their infrastructure, telephone companies use one of those hierarchical systems that is made up of groups, supergroups, master groups, and jumbo groups



In this **analog hierarchy,** 12 voice channels are multiplexed onto a higher-bandwidth line to create a **group.** A group has 48 kHz of bandwidth and supports 12 voice channels.

At the next level, up to five groups can be multiplexed to create a composite signal called a **supergroup.** A supergroup has a bandwidth of 240 kHz and supports up to 60 voice channels. Supergroups can be made up of either five groups or 60 independent voice channels.

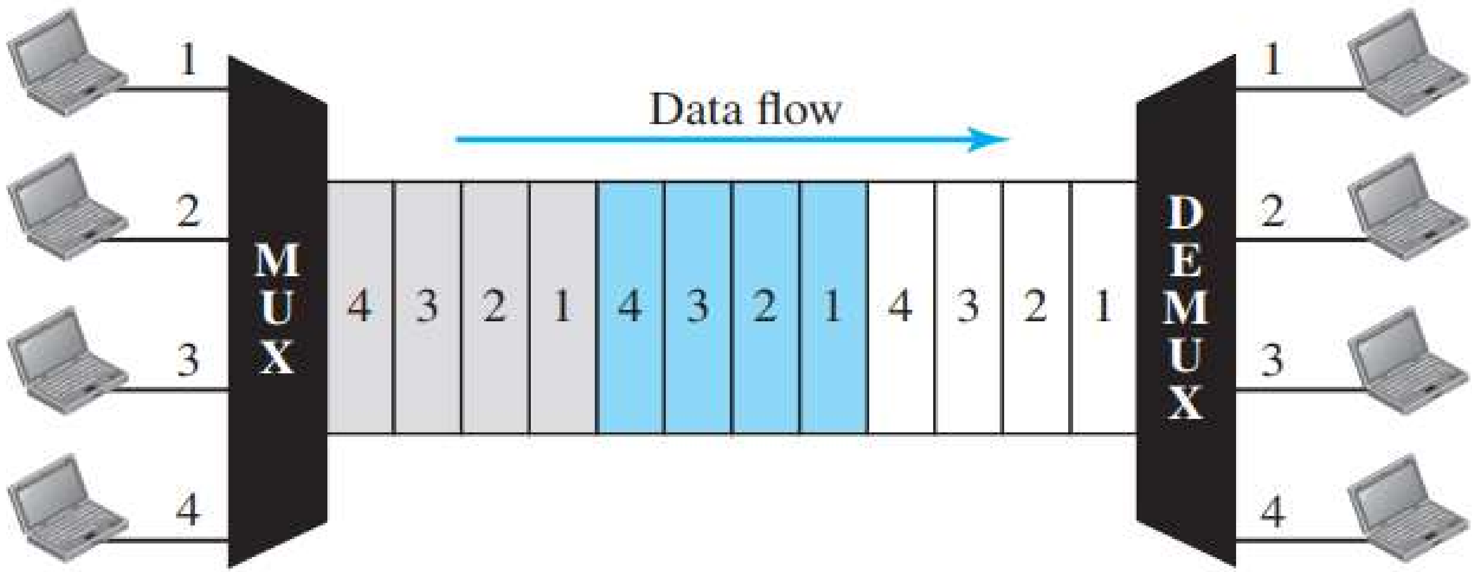
At the next level, 10 supergroups are multiplexed to create a **master group.** A master group must have 2.40 MHz of bandwidth, but the need for guard bands between the supergroups increases the necessary bandwidth to 2.52 MHz. Master groups support up to 600 voice channels.

Finally, six master groups can be combined into a **jumbo group.** A jumbo group must have 15.12 MHz (6×2.52 MHz) but is augmented to 16.984 MHz to allow for guard bands between the master groups.

Time-Division Multiplexing

- Time-division multiplexing (TDM) is a digital process that allows several connections to share the high bandwidth of a link.
- Instead of sharing a portion of the bandwidth as in FDM, time is shared.
- Each connection occupies a portion of time in the link.
- Digital data from different sources are combined into one timeshared link.

- TDM is divided into two different schemes: **Synchronous** and **Statistical**.

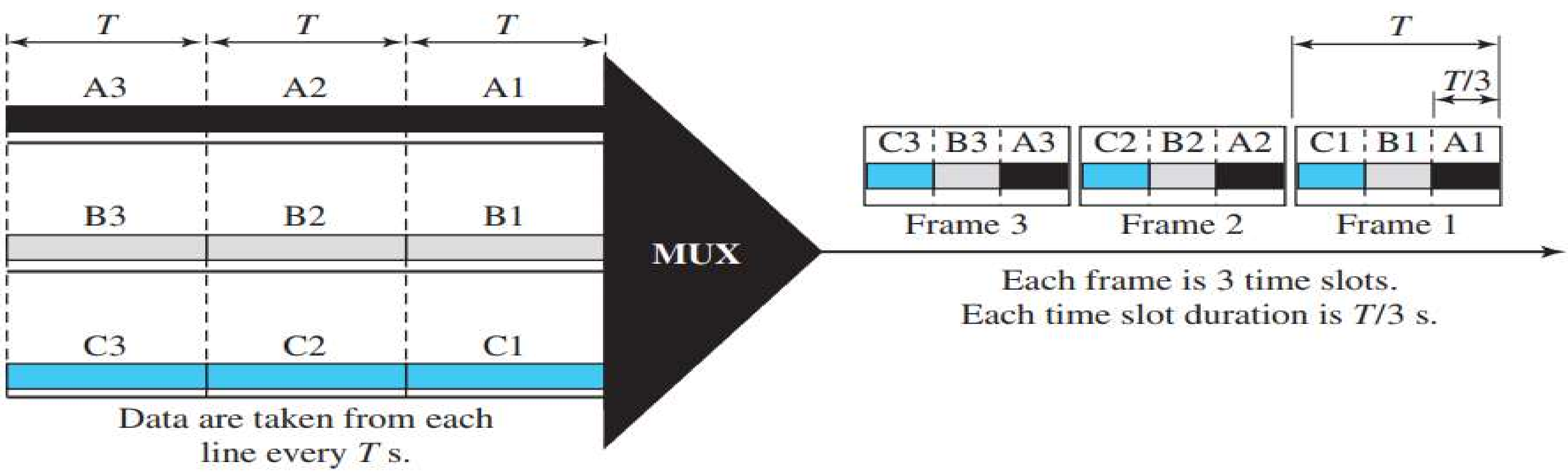


Synchronous TDM

In synchronous TDM, each input connection has an allotment in the output even if it is not sending data.

- In synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one input time slot.
- A unit can be 1 bit, one character, or one block of data.
- Each input unit becomes one output unit and occupies one output time slot. However, the duration of an output time slot is **n** times shorter than the duration of an input time slot.
- If an input time slot is **T** s, the output time slot is **T/n** s, where n is the number of connections.
- In other words, a unit in the output connection has a shorter duration; it travels faster.

- If we have **n** connections, a frame is divided into **n** time slots and one slot is allocated for each unit, one for each input line.
- If the duration of the input unit is **T**, the duration of each slot is **T/n** and the duration of each frame is **T**.
- The data rate of the output link must be **n** times the data rate of a connection to guarantee the flow of data.



In above figure, the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of

1. each input slot,
2. each output slot, and
3. each frame?

Solution

We can answer the questions as follows:

1. The data rate of each input connection is 1 kbps. This means that the bit duration is 1/1000 s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).
2. The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is 1/3 ms.
3. Each frame carries three output time slots. So the duration of a frame is $3 \times 1/3$ ms, or 1 ms. The duration of a frame is the same as the duration of an input unit.

Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (1) the duration of 1 bit before multiplexing, (2) the transmission rate of the link, (3) the duration of a time slot, and (4) the duration of a frame.

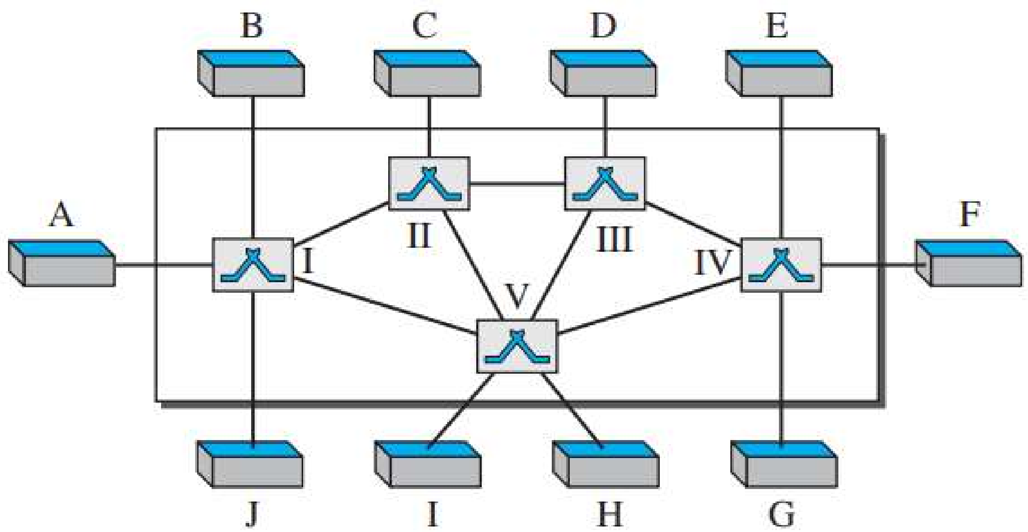
Solution

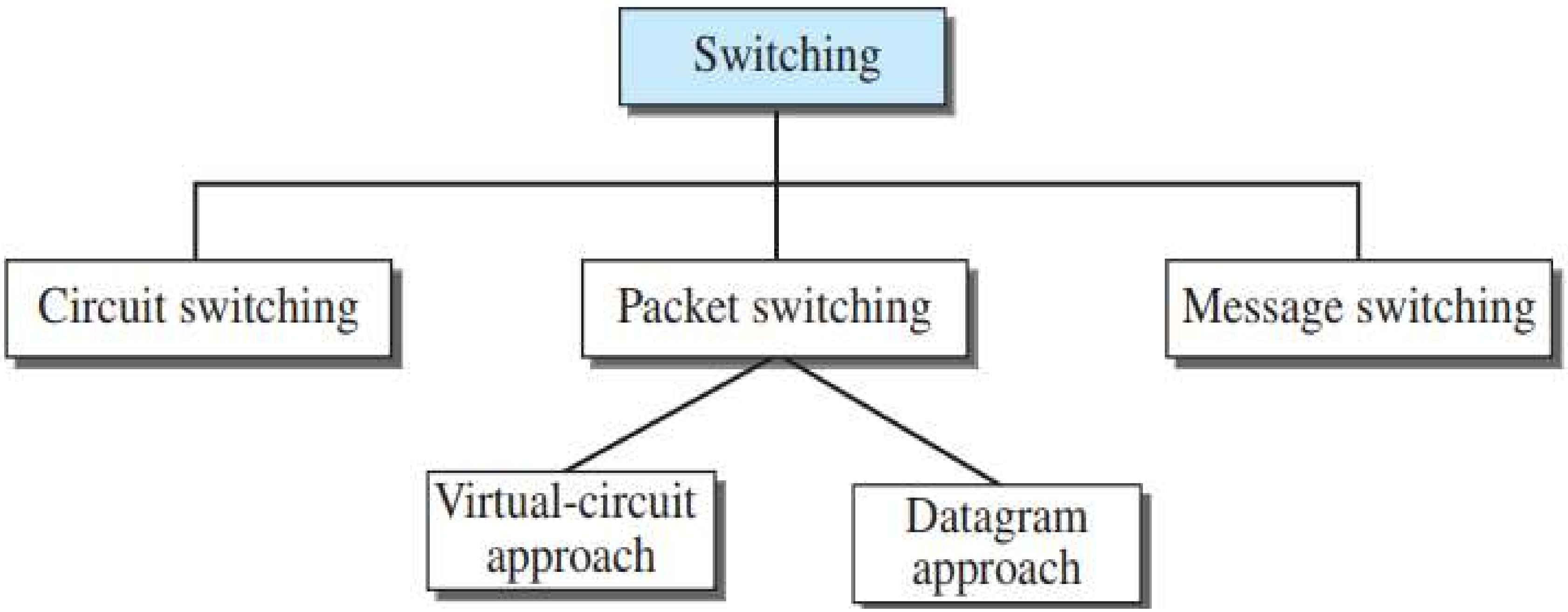
We can answer the questions as follows:

1. The duration of 1 bit before multiplexing is 1/1 kbps, or 0.001 s (1 ms).
2. The rate of the link is 4 times the rate of a connection, or 4 kbps.
3. The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or 1/4 ms or 250 μs. Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or 1/4 kbps or 250 μs.
4. The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms. We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times 250 μs, or 1 ms.

Switching

- A switched network consists of a series of interlinked nodes, called switches.
- Switches are devices capable of creating temporary connections between two or more devices linked to the switch.
- In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only for routing.




Switching
Circuit switching
Packet switching
Message switching
Virtual-circuit approach
Datagram approach

Switching can happen at several layers of the TCP/IP protocol suite.

Switching at Physical Layer

At the physical layer, we can have only circuit switching. There are no packets exchanged at the physical layer. The switches at the physical layer allow signals to travel in one path or another.

Switching at Data-Link Layer

At the data-link layer, we can have packet switching. However, the term *packet* in this case means *frames* or *cells*. Packet switching at the data-link layer is normally done using a virtual-circuit approach.

Switching at Network Layer

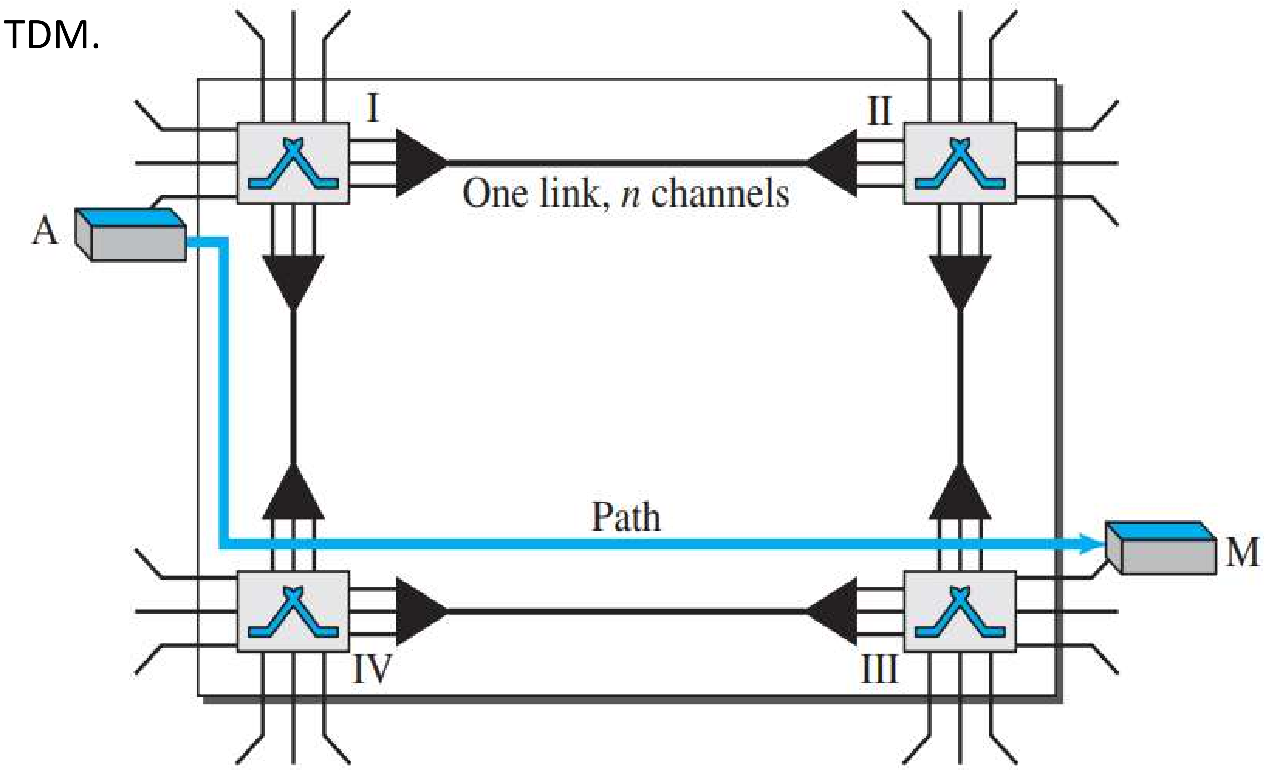
At the network layer, we can have packet switching. In this case, either a virtual-circuit approach or a datagram approach can be used. Currently the Internet uses a datagram approach

Switching at Application Layer

At the application layer, we can have only message switching. The communication at the application layer occurs by exchanging messages. Conceptually, we can say that communication using e-mail is a kind of message-switched communication, but we do not see any network that actually can be called a message-switched network.

CIRCUIT-SWITCHED NETWORKS

- A circuit-switched network consists of a set of switches connected by **physical links**.
- A connection between two stations is a **dedicated path** made of one or more links. However, each connection uses only **one dedicated channel on each link**. Each link is normally divided into n channels by using FDM or TDM.



The actual communication in a circuit-switched network requires three phases:

- Connection setup
- Data transfer
- Connection teardown

Setup Phase

- Before the two parties (or multiple parties in a conference call) can communicate, a dedicated circuit (combination of channels in links) needs to be established.
- The end systems are normally connected through dedicated lines to the switches, so connection setup means creating dedicated channels between the switches.
  - ❖ For example, in above figure, when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I.
  - ❖ Switch I finds a channel between itself and switch IV that can be dedicated for this purpose.
  - ❖ Switch I then sends the request to switch IV, which finds a dedicated channel between itself and switch III.
  - ❖ Switch III informs system M of system A's intention at this time.

- In the next step to making a connection, an acknowledgment from system M needs to be sent in the opposite direction to system A. Only after system A receives this acknowledgment is the connection established.
- Note that end-to-end addressing is required for creating a connection between the two end systems.

Data-Transfer Phase

- After the establishment of the dedicated circuit (channels), the two parties can transfer data.

Teardown Phase

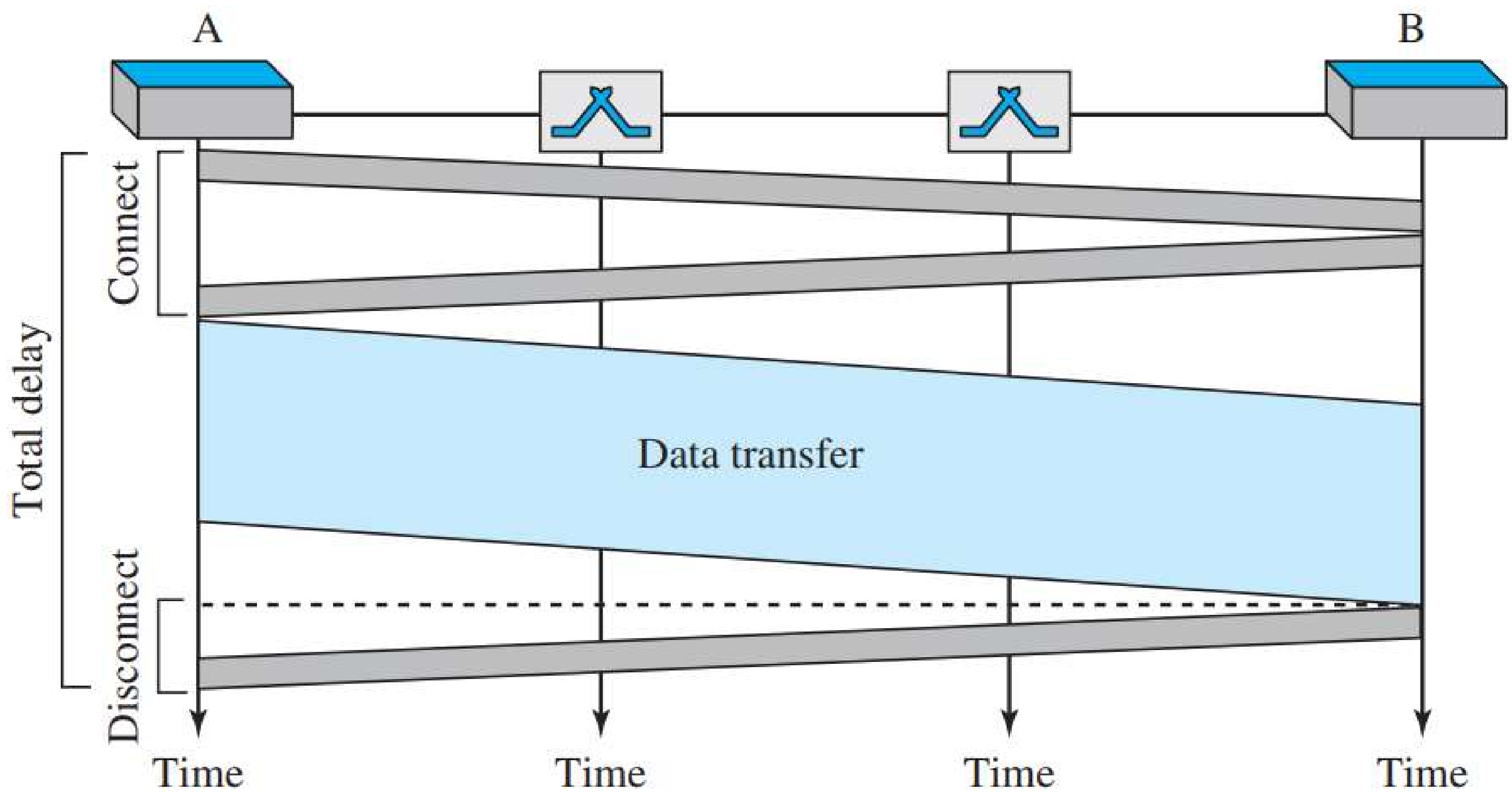
- When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

Efficiency

- It can be argued that circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection.
- These resources are unavailable to other connections.
- Allowing resources to be dedicated means that other connections are deprived.

Delay

- Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal.
- During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection.


A
B
Total delay
Connect
Data transfer
Disconnect
Time
Time
Time
Time

- The total delay is due to the time needed to create the **Connection**, **Transfer Data**, and **Disconnect the Circuit**.
- The **delay caused by the setup** is the sum of four parts:
  - ✓ The propagation time of the source computer request (slope of the first gray box)
  - ✓ The request signal transfer time (height of the first gray box)
  - ✓ The propagation time of the acknowledgment from the destination computer (slope of the second gray box)
  - ✓ The signal transfer time of the acknowledgment (height of the second gray box).
- The **delay due to data transfer** is the sum of two parts:
  - ✓ The propagation time (slope of the colored box) and data transfer time (height of the colored box).
- The third box shows the time needed to tear down the circuit. The case in which the receiver requests disconnection, which creates the maximum delay is shown here.

PACKET SWITCHING

- In **packet-switched network**, the message needs to be **divided into packets** of fixed or variable size.
- The size of the packet is determined by the network and the governing protocol.
- In packet switching, there is no resource allocation for a packet.
- This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet.
- **Resources are allocated on demand and done on a first-come, first-served basis**.
- When a switch receives a packet, no matter what the source or destination is, the packet must wait if there are other packets being processed.

There are two types of packet-switched networks:

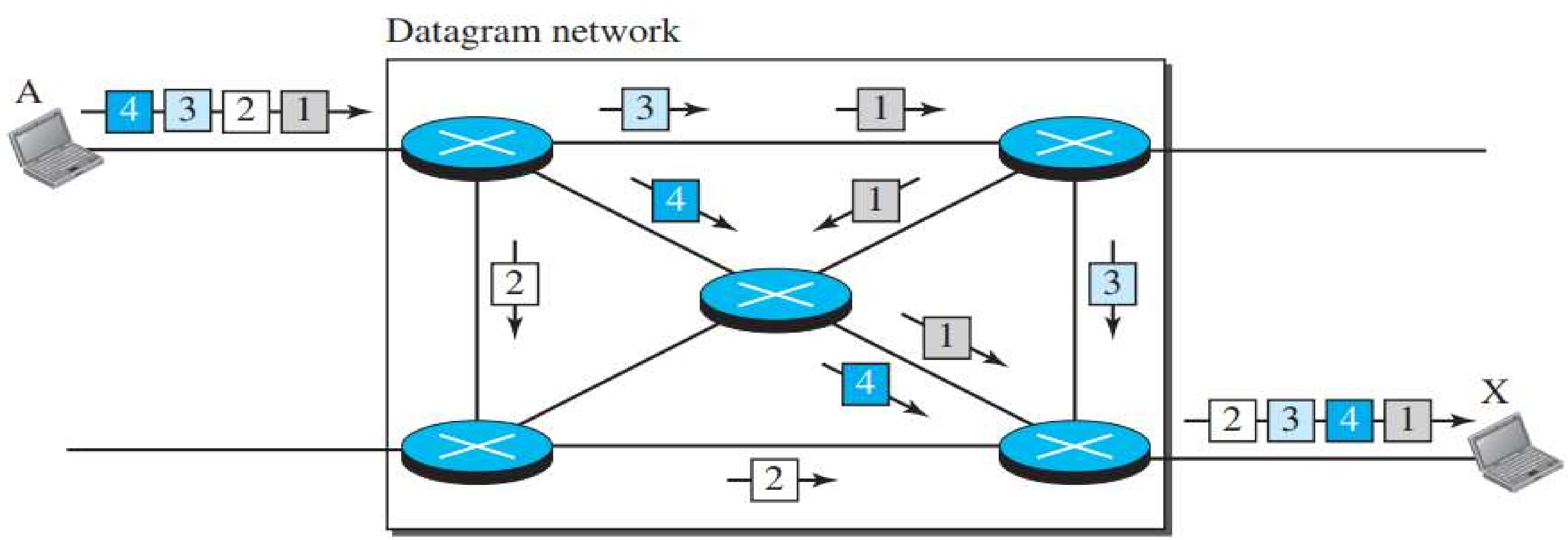
- Datagram networks
- Virtual-Circuit networks.

Datagram Networks

- In a datagram network, each packet is treated independently of all others.
- Even if a packet is part of a multi-packet transmission, the network treats it as though it existed alone.
- Packets in this approach are referred to as **Datagrams**.
- Datagram switching is normally done at the network layer.

- In the given example, all four packets (or datagrams) belong to the same message, but may travel different paths to reach their destination.
- This is so because the links may be involved in carrying packets from other sources and do not have the necessary bandwidth available to carry all the packets from A to X.
- This approach can cause the datagrams of a transmission to arrive at their destination out of order with different delays between the packets.
- Packets may also be lost or dropped because of a lack of resources.

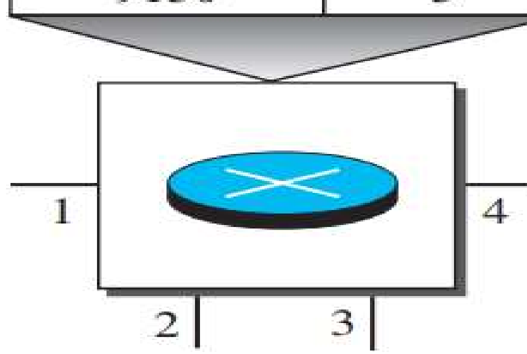
- In most protocols, it is the responsibility of an upper-layer protocol to reorder the datagrams or ask for lost datagrams before passing them on to the application.
- The datagram networks are sometimes referred to as **Connectionless networks**.
- The term **connectionless** here means that the switch (packet switch) does not keep information about the connection state.
- There are no setup or teardown phases.
- Each packet is treated the same by a switch regardless of its source or destination.



Routing Table

- In this type of network, **each switch** (or packet switch) has a **routing table** which is based on the destination address.
- The routing tables are dynamic and are updated periodically.
- The destination addresses and the corresponding forwarding output ports are recorded in the tables.

Destination address	Output port
1232	1
4150	2
⋮	⋮
9130	3



Destination Address

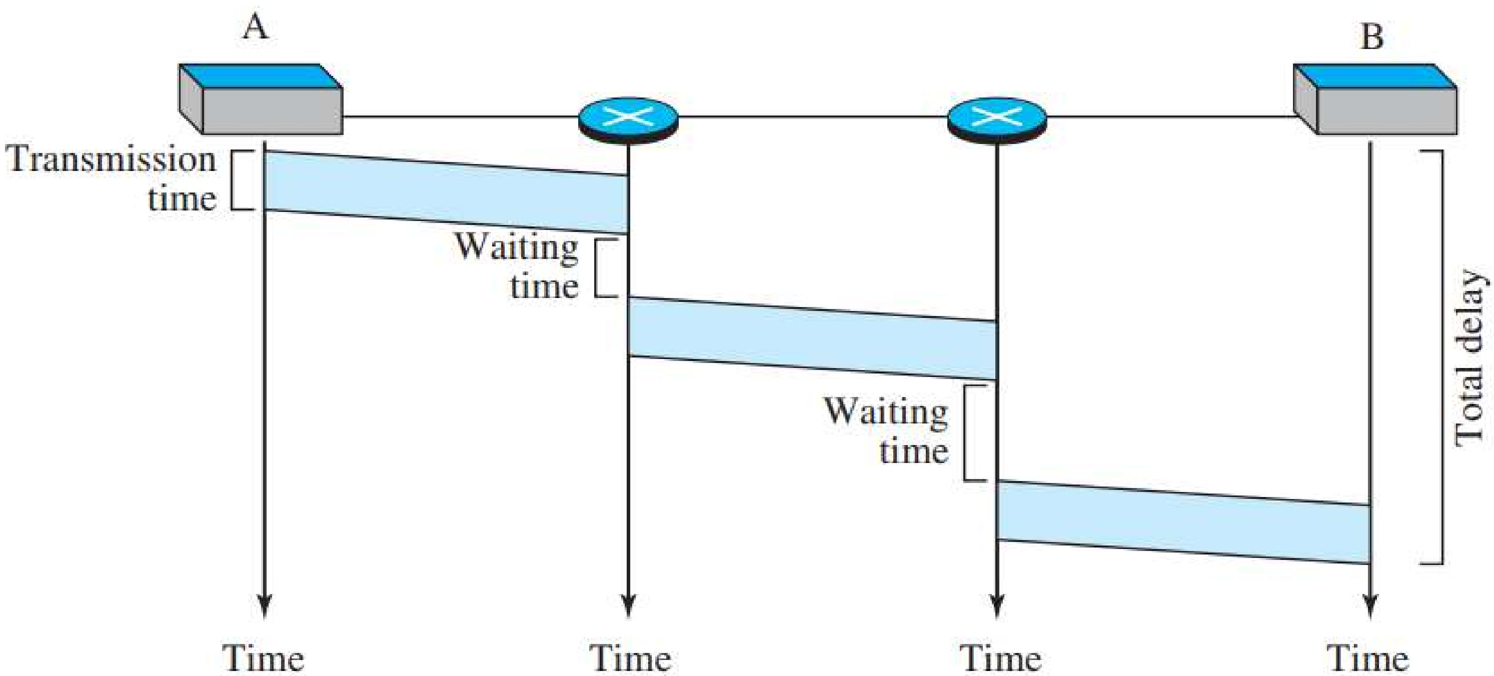
- Every packet in a datagram network carries a header that contains, among other information, the destination address of the packet.
- When the switch receives the packet, this destination address is examined; the routing table is consulted to find the corresponding port through which the packet should be forwarded.
- This address remains the same during the entire journey of the packet.

Efficiency

- The efficiency of a datagram network is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred.
- If a source sends a packet and there is a delay of a few minutes before another packet can be sent, the resources can be reallocated during these minutes for other packets from other sources.

Delay

- There may be greater delay in a datagram network than in a virtual-circuit network.
- Although there are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded.
- In addition, since **not all packets in a message necessarily travel through the same switches**, the delay is not uniform for the packets of a message.



The packet travels through two switches. There are three transmission times (3*T*), three propagation delays (slopes 3τ of the lines), and two waiting times (*w*1 + *w*2). We ignore the processing time in each switch. The total delay is

$$\text{Total delay} = 3T + 3\tau + w_1 + w_2$$

Virtual-Circuit Network

To be done by student